

Genomic null calibration: interim results

2026-10-07 | Frozen snapshot: **535 / 6,000** completed simulations (8.9%). Target: 1,000 per population. Live runner at report generation: running; 8 workers; 0 reported failures.

Decoded TMRCA: all sampled haplotype pairs

Four requested numbers per population. Entries are percentages of pairs with TMRCA below the specified age; they are descriptive empirical 95th/99th percentiles across retained null focal sites.

Population	Nulls	10 kya: top 5%	10 kya: top 1%	50 kya: top 5%	50 kya: top 1%
AFR	88	1.13%	1.50%	8.27%	12.14%
AMR	90	2.14%	2.51%	24.44%	32.74%
EAS	89	5.18%	6.36%	31.42%	42.28%
EUR	90	3.86%	5.64%	29.54%	45.59%
MID	89	2.69%	3.17%	26.89%	32.31%
SAS	89	3.45%	4.34%	24.91%	37.19%

These percentiles are provisional

Fewer than 99 nulls are available for AFR, AMR, EAS, EUR, MID, SAS, so **p <= 0.01 cannot yet be attained** for those populations under $p = (1 + \text{number of null scores at least as large as the target}) / (n + 1)$. Their displayed top-1% percentile is the sample maximum, not a validated $p = 0.01$ threshold.

Pooled CDFs are shown without demographic confidence bands. Every represented history currently has one completed region; ten are planned per history. Completed runs may favor faster histories.

Exact p-value boundaries and interpretation

For the decoded all-pairs statistic, reject at $p \leq 0.05$ only when the target score is **strictly greater** than the boundary below. Equality does not reject. These boundaries can differ from descriptive percentiles.

Population	10 kya boundary	50 kya boundary	Smallest possible p
AFR	1.25%	8.55%	0.0112
AMR	2.23%	25.69%	0.0110
EAS	5.63%	31.60%	0.0111
EUR	3.98%	35.23%	0.0110
MID	2.76%	27.05%	0.0111
SAS	3.61%	27.43%	0.0111

True TMRCA: descriptive percentiles for all pairs

Population	Nulls	10 kya: top 5%	10 kya: top 1%	50 kya: top 5%	50 kya: top 1%
AFR	88	0.49%	0.56%	5.26%	5.74%
AMR	90	0.65%	0.75%	14.77%	17.16%
EAS	89	0.99%	1.06%	26.56%	29.72%
EUR	90	1.35%	1.59%	25.74%	31.44%
MID	89	0.68%	0.74%	17.50%	26.06%
SAS	89	0.89%	1.22%	23.62%	29.66%

Model, allele frequencies and limitations

Neutral archaic-derived alleles, segregating immediately after 2% introgression at 50 kya and observed in the present sample (fixation retained). Neanderthal $N_e = 3,600$; split = 500 kya; 25 years/generation. 200 sampled diploids; 10,000 fixed sampled haplotype pairs; no rescaling.

Random autosomal regions use the local deCODE GRCh38 map in simulation. Gamma-SMC uses the cropped region mean recombination rate, accurate exp10, and fixed backward alignment. Mutation rate = $1.29e-8$. Assembly gaps and missing map coverage are masked; the HMMix strict mask is not used.

True versus decoded and ALT/ALT results

The following pages include allele-frequency distributions, separate true/decoded CDFs, and all-pairs/ALT-ALT comparisons. ALT/ALT scores use raw carrier-pair fractions. Undefined carrier-pair scores are excluded from displayed CDFs and descriptive percentiles; exact tests retain them in the null denominator as no-calls. See the CSVs for availability and saturation flags.

This campaign contains calibration nulls only. It does not supply new selected-treatment power or independent neutral-target FPR estimates.

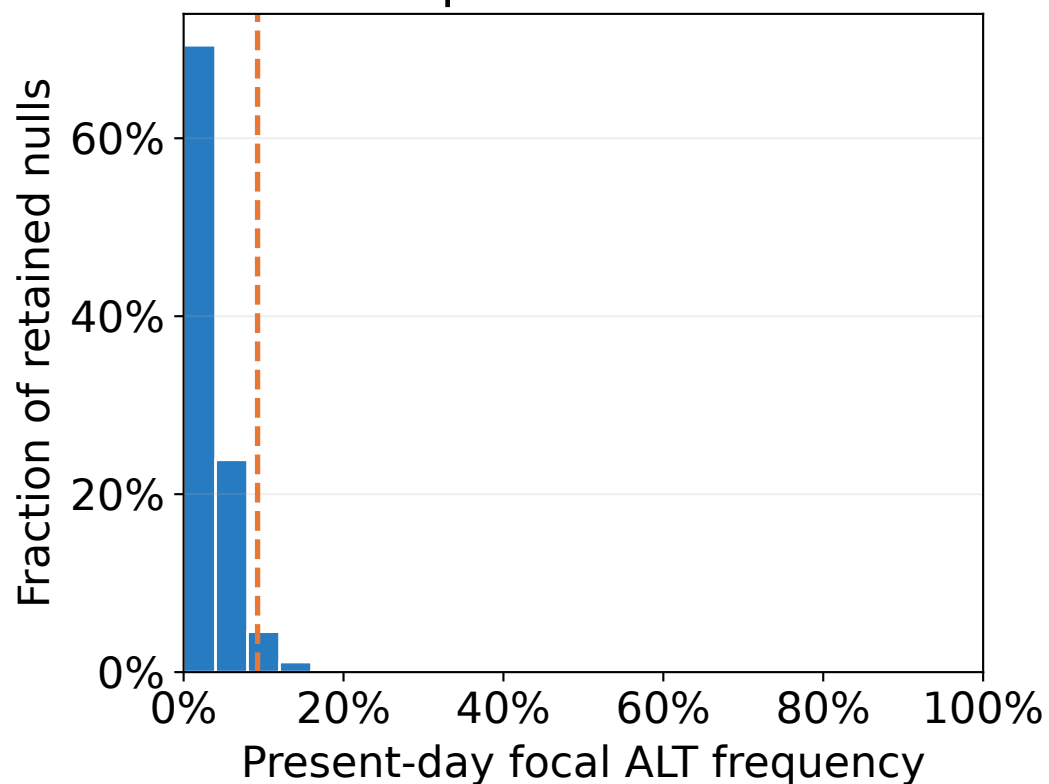
ALT/ALT at 50 kya is saturated: every population has a 100% upper-tail boundary for both true and decoded TMRCA. Even a target score of 100% cannot reject at $p \leq 0.05$ with these nulls.

Focal ALT frequencies among retained nulls

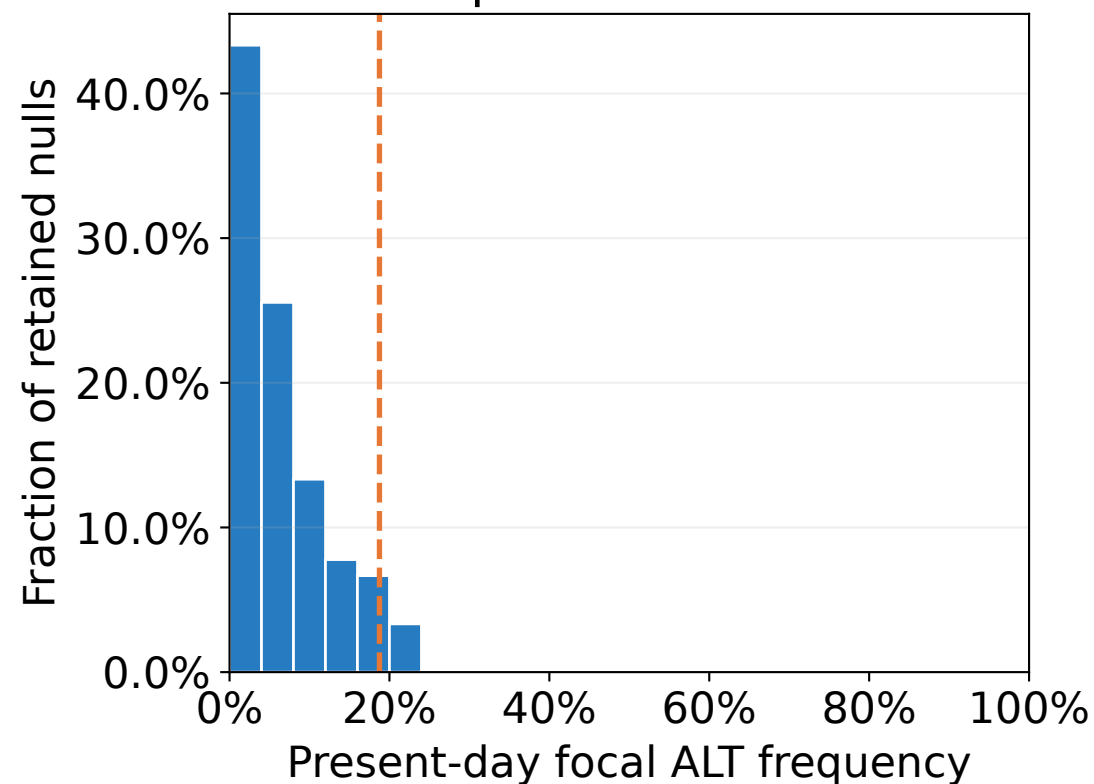
Population	Median AF	95th-percentile AF	Fixed alleles	Median attempts
AFR	2.0%	9.2%	0	7
AMR	5.3%	18.8%	0	12
EAS	6.2%	25.5%	0	17
EUR	7.8%	28.0%	0	13.5
MID	5.2%	20.8%	0	12
SAS	7.2%	30.0%	0	15

----- Empirical 95th percentile

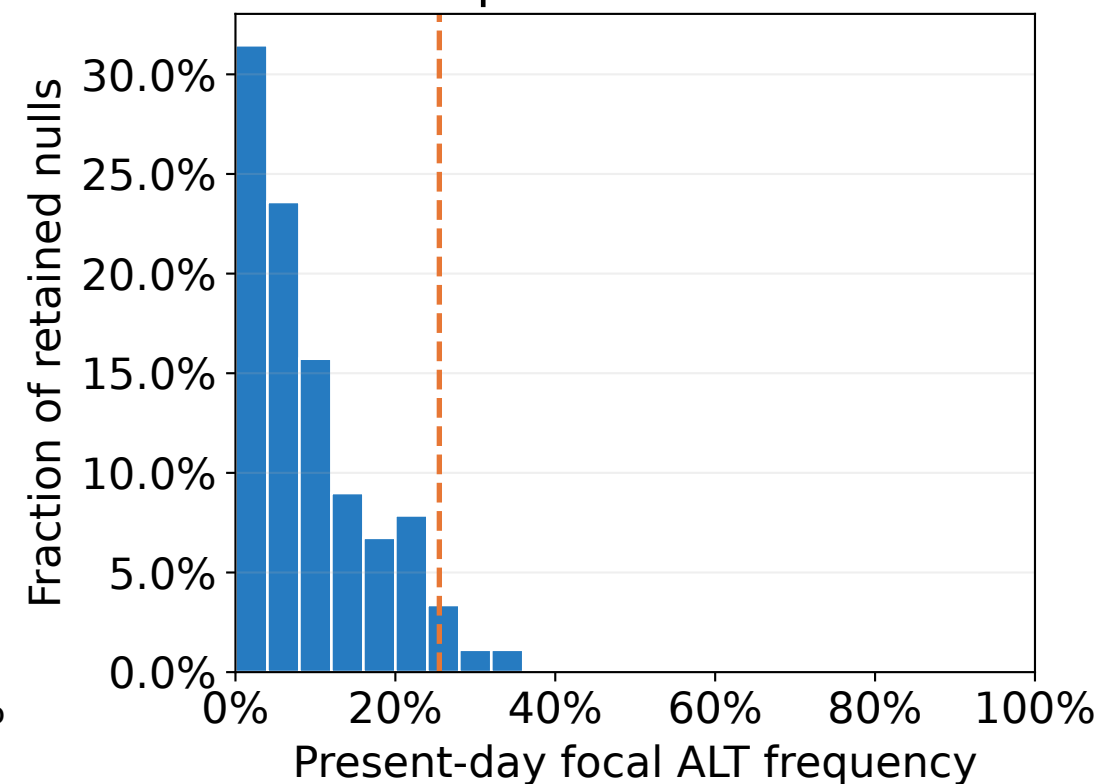
AFR | n = 88
95th percentile: 9.2%



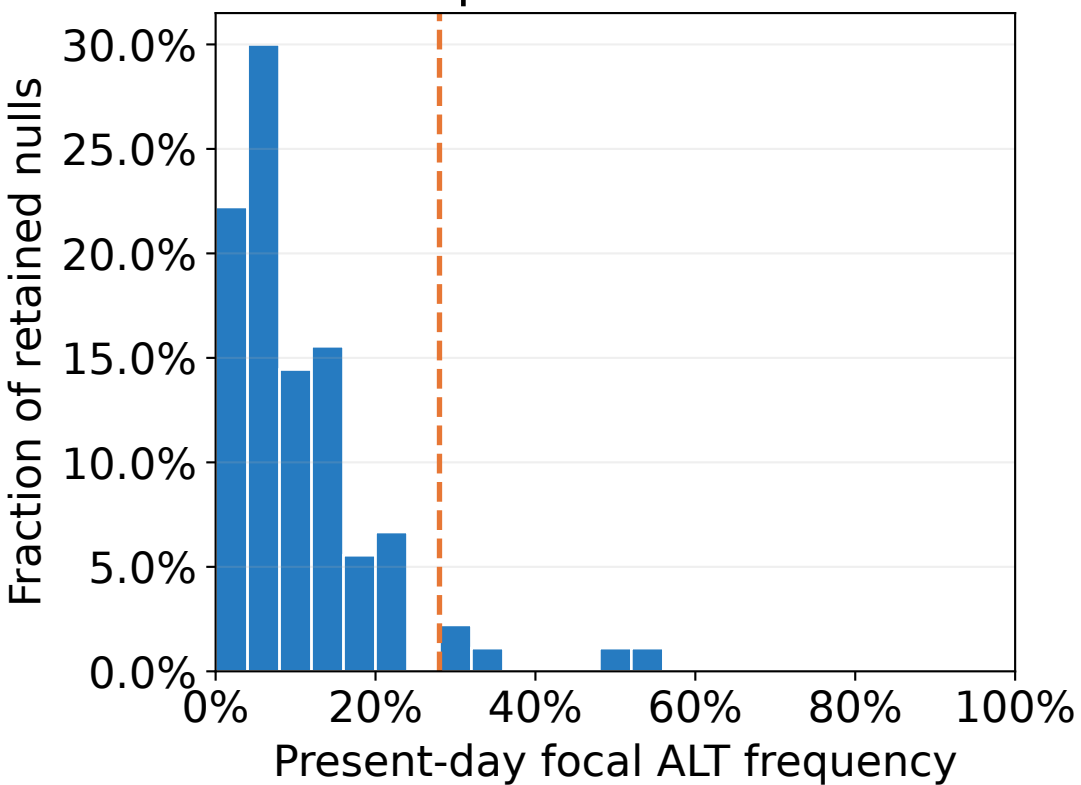
AMR | n = 90
95th percentile: 18.8%



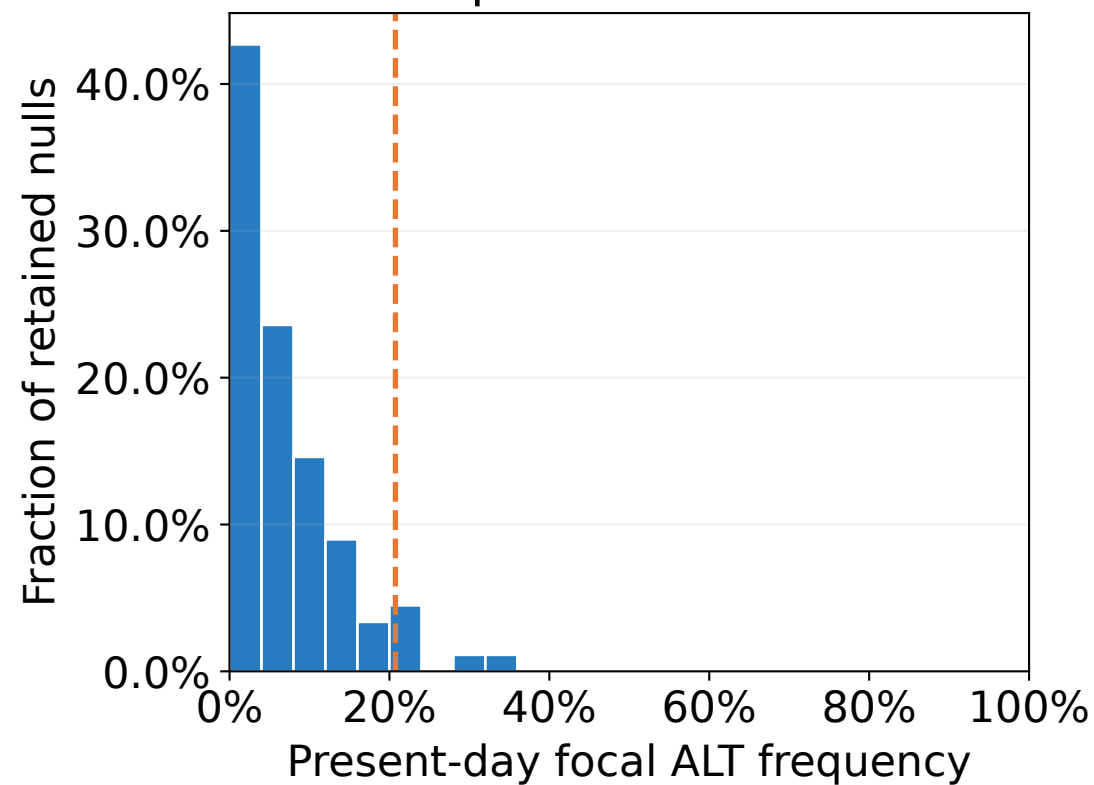
EAS | n = 89
95th percentile: 25.5%



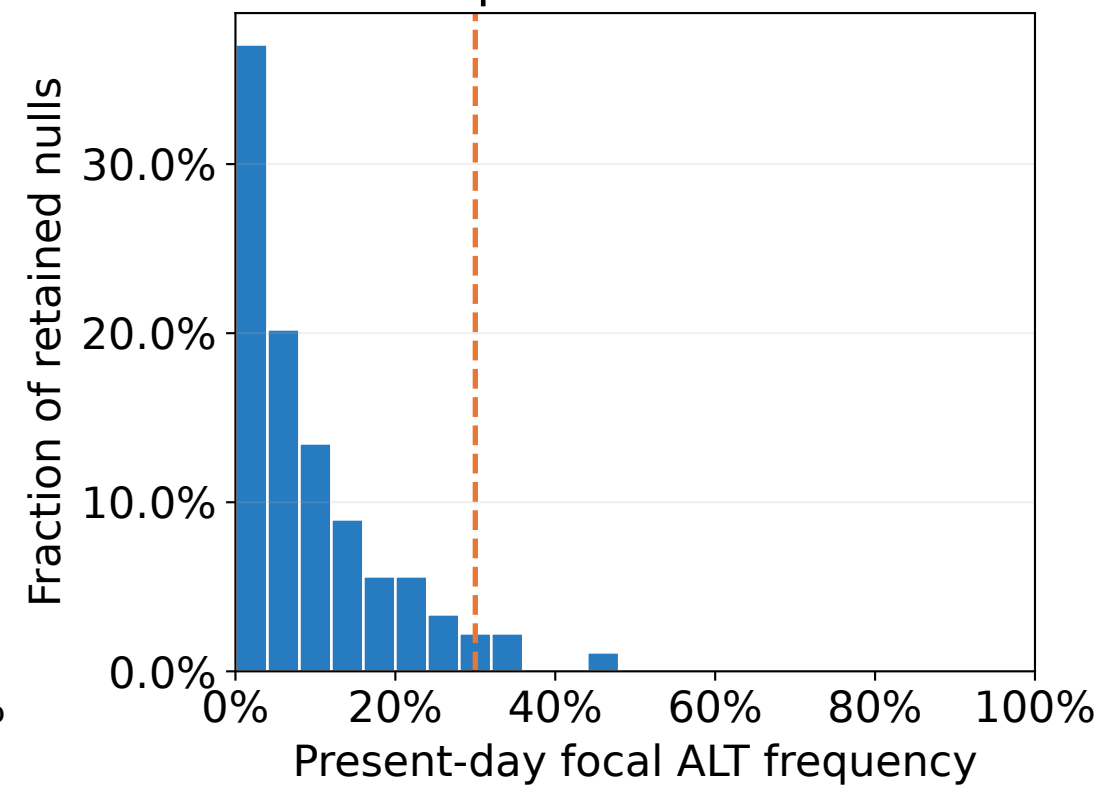
EUR | n = 90
95th percentile: 28.0%



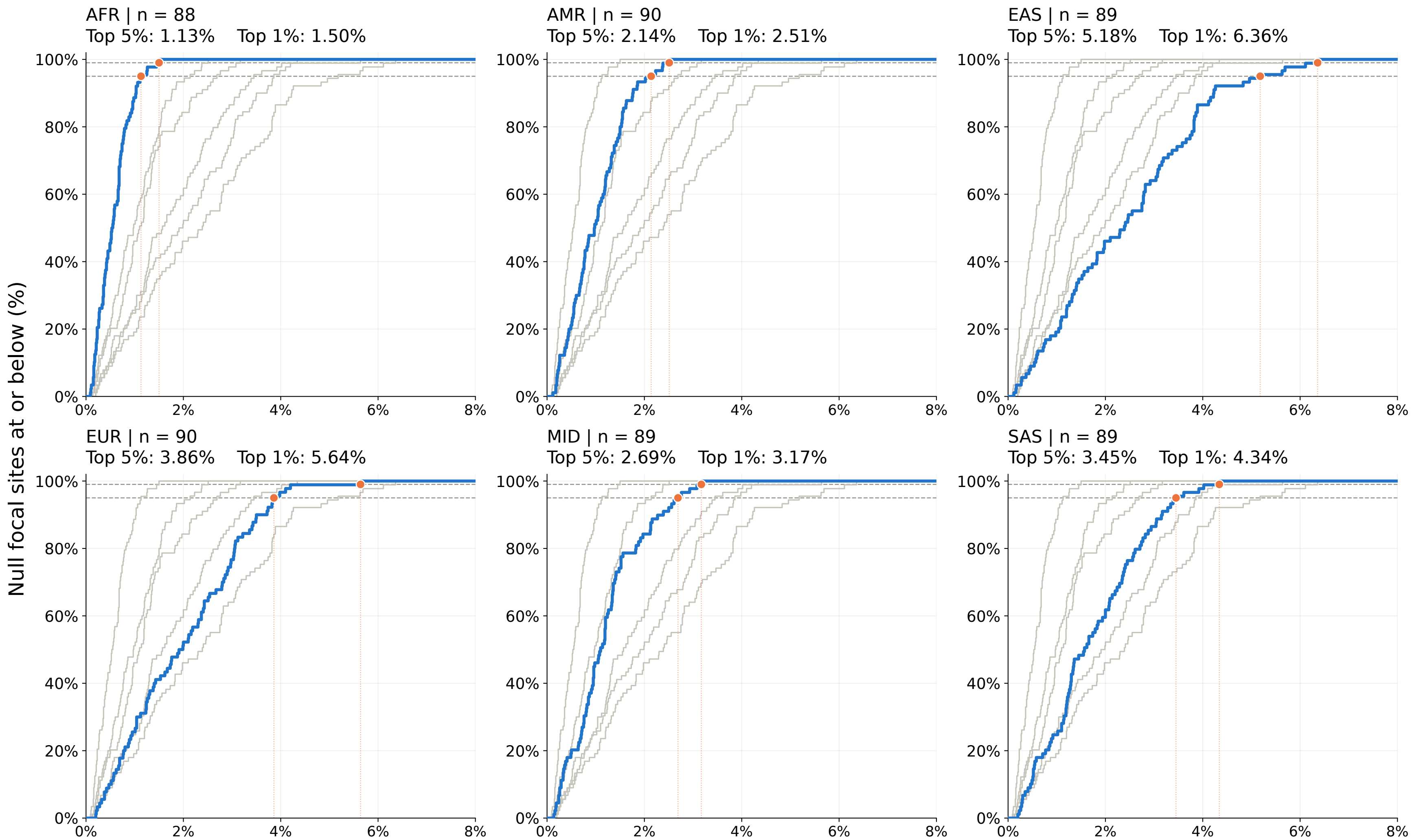
MID | n = 89
95th percentile: 20.8%



SAS | n = 89
95th percentile: 30.0%



Decoded TMRCA | All sampled pairs | T < 10,000 years

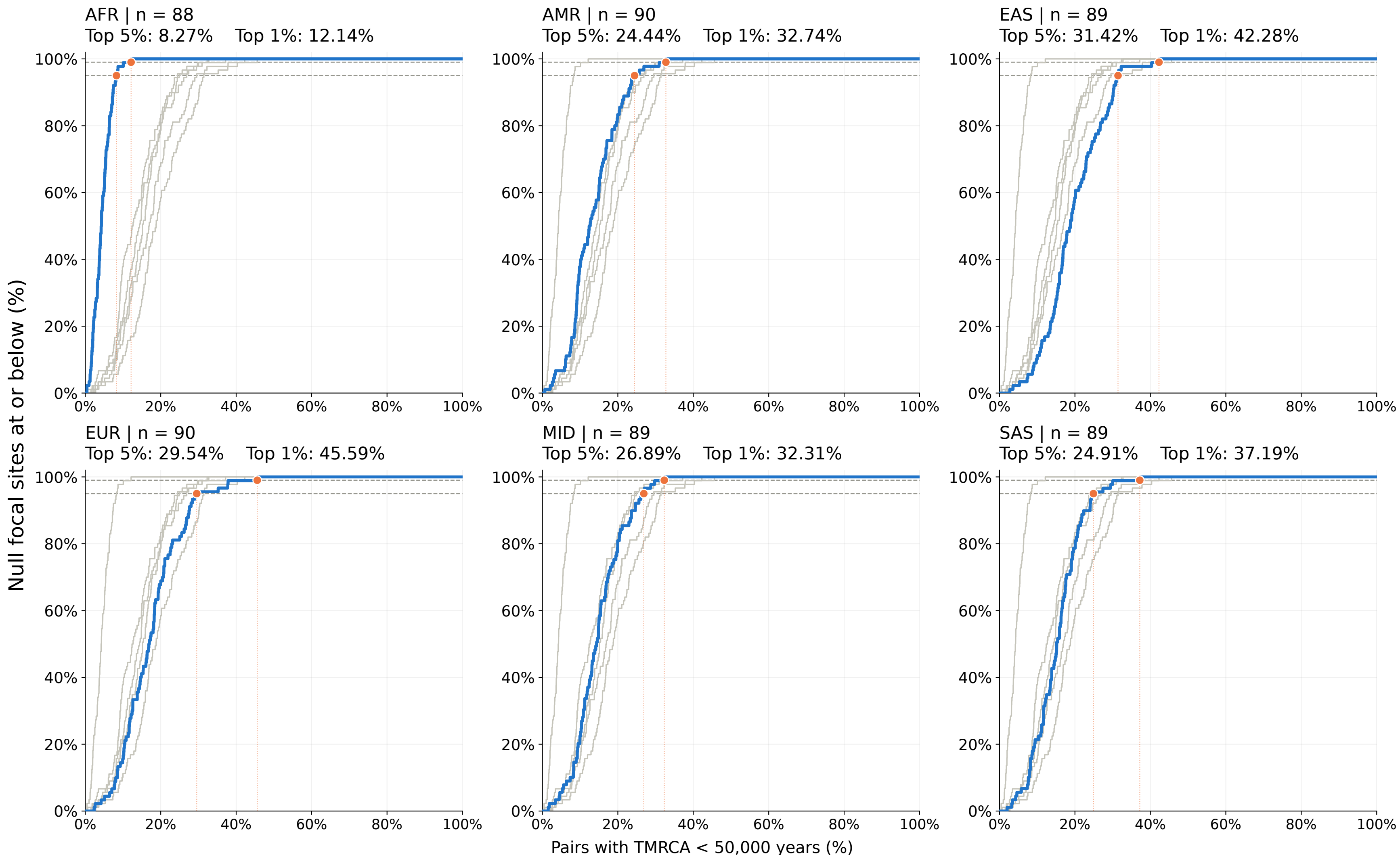


Pairs with TMRCA < 10,000 years (%)

PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

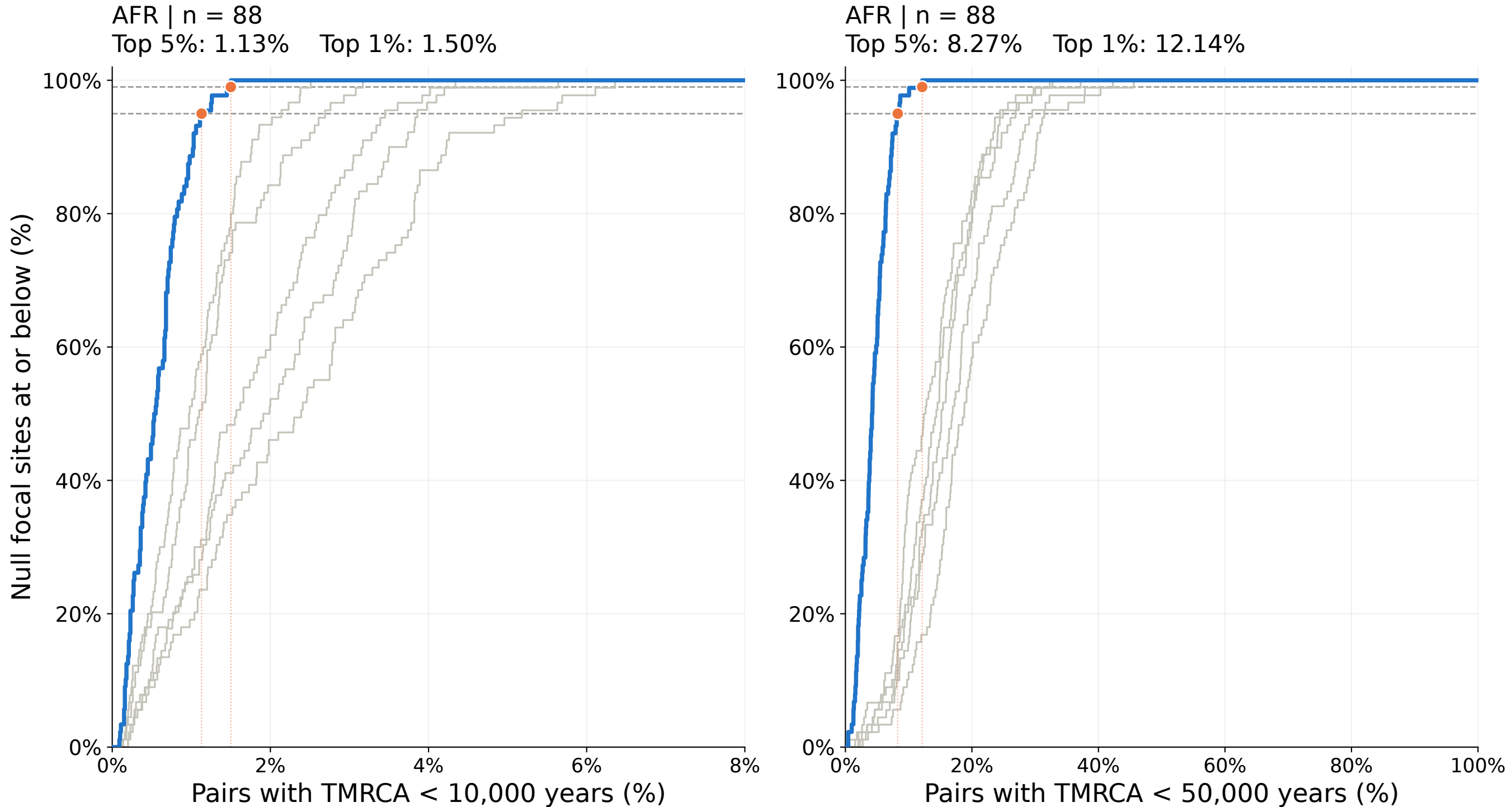
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | All sampled pairs | T < 50,000 years



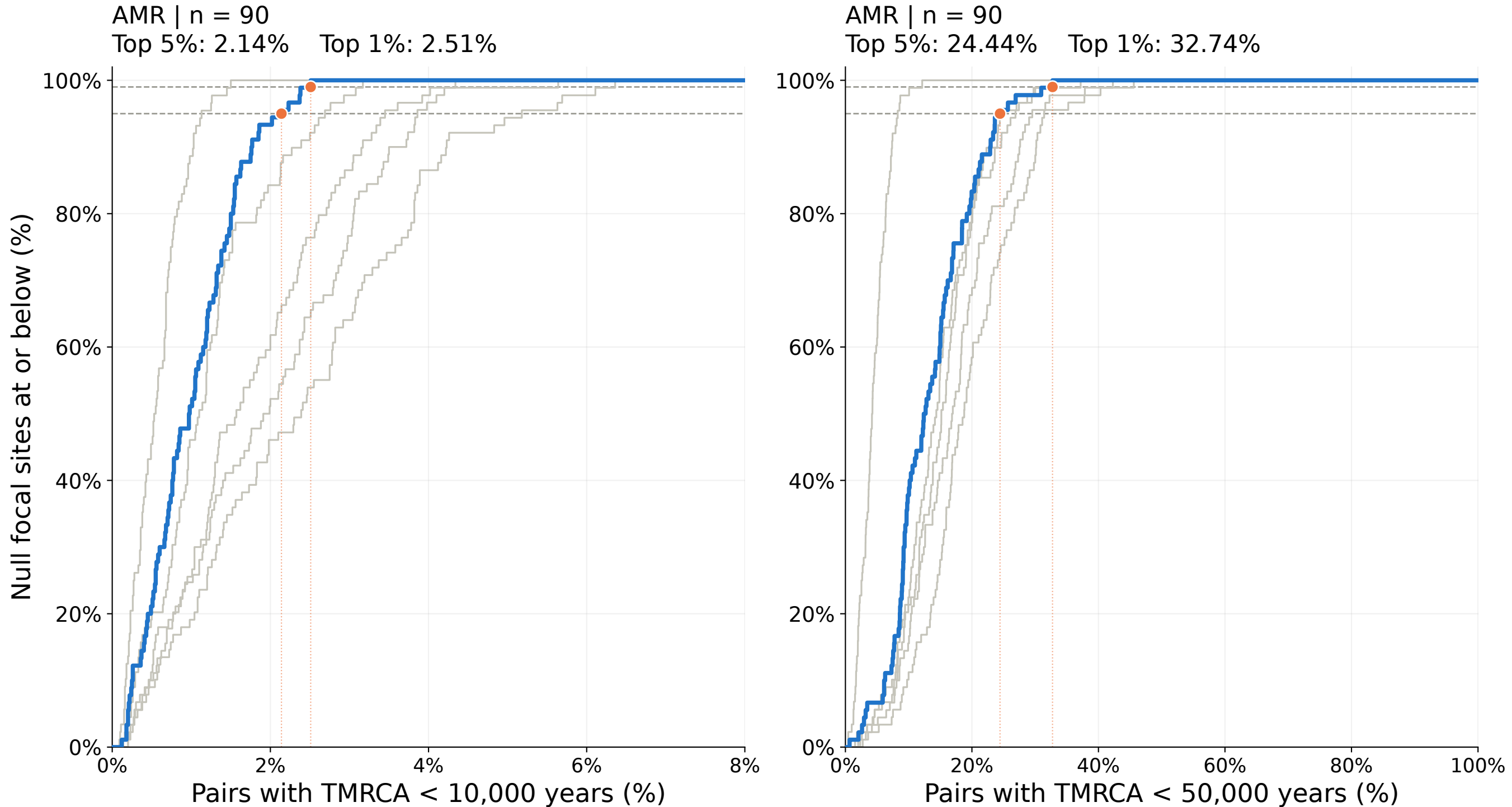
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | All sampled pairs



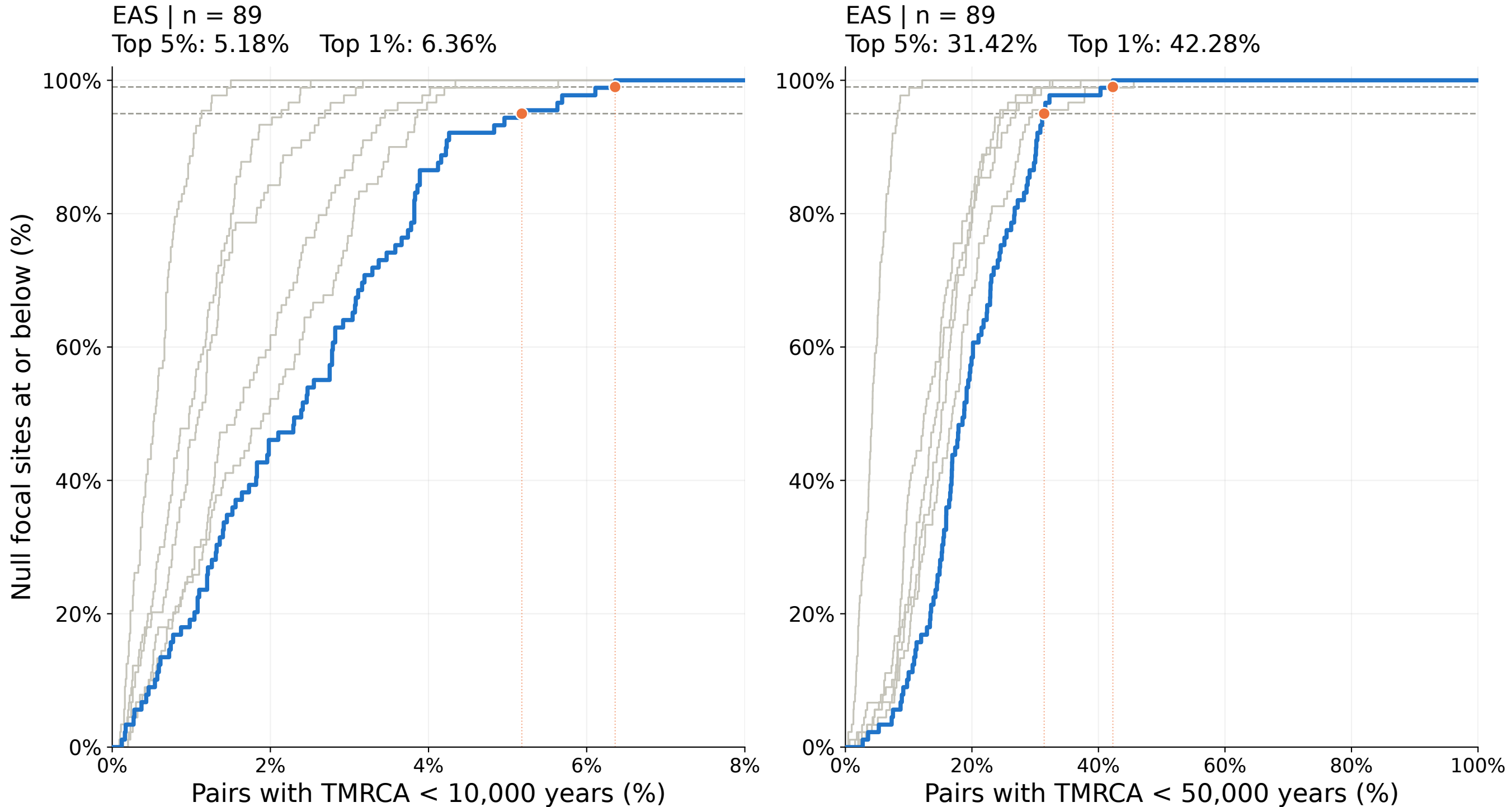
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | All sampled pairs



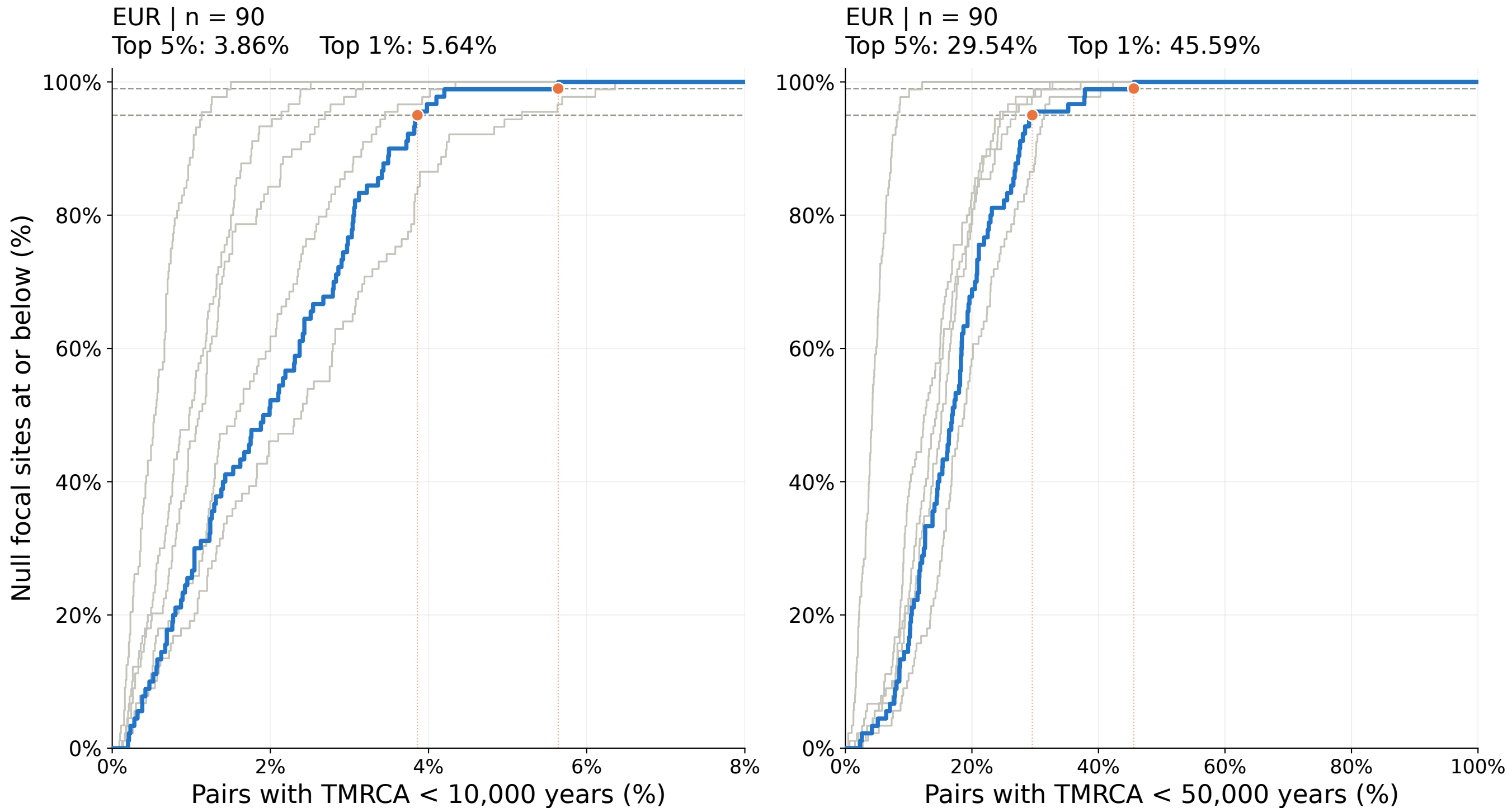
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | All sampled pairs



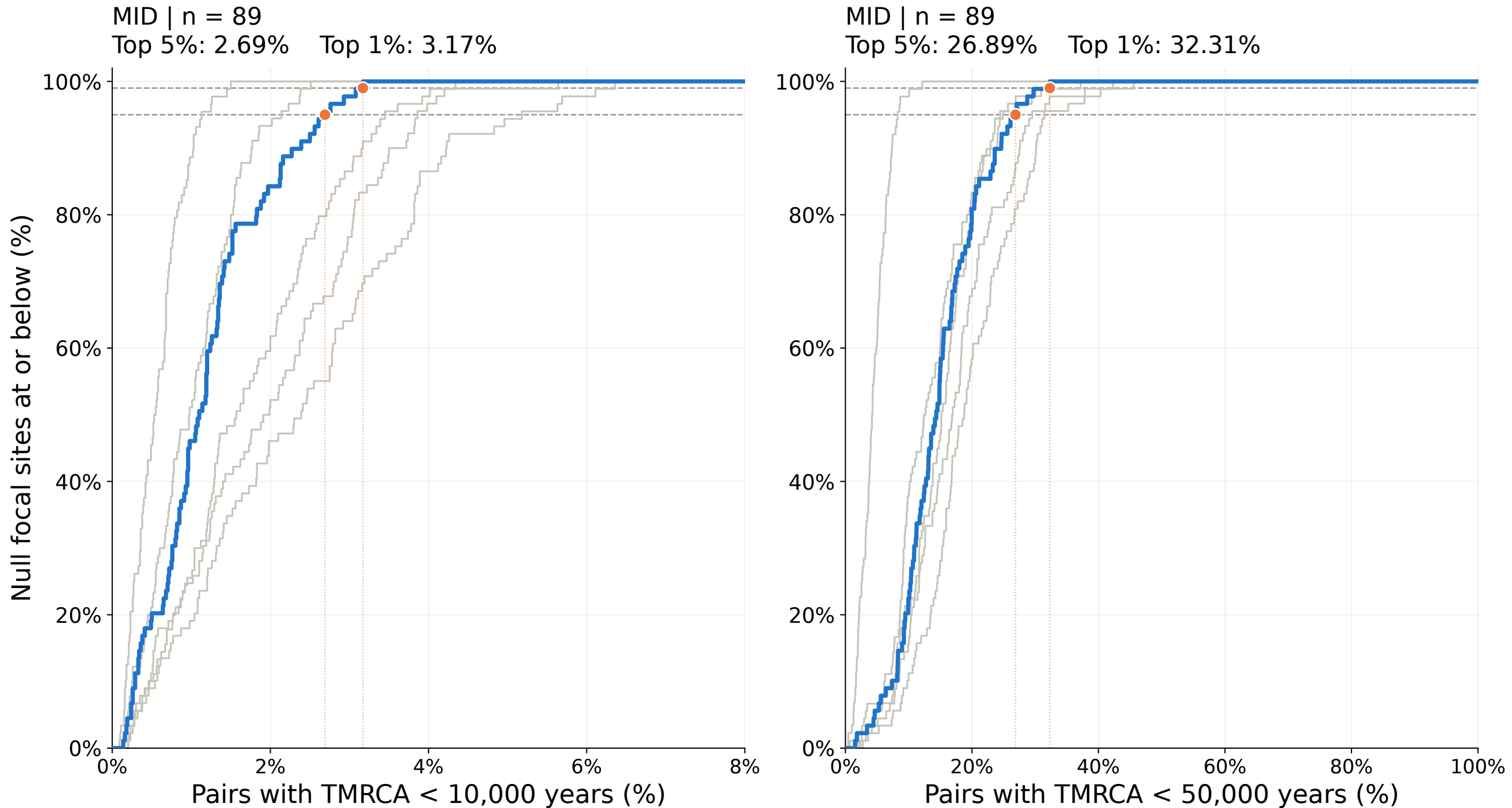
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | All sampled pairs



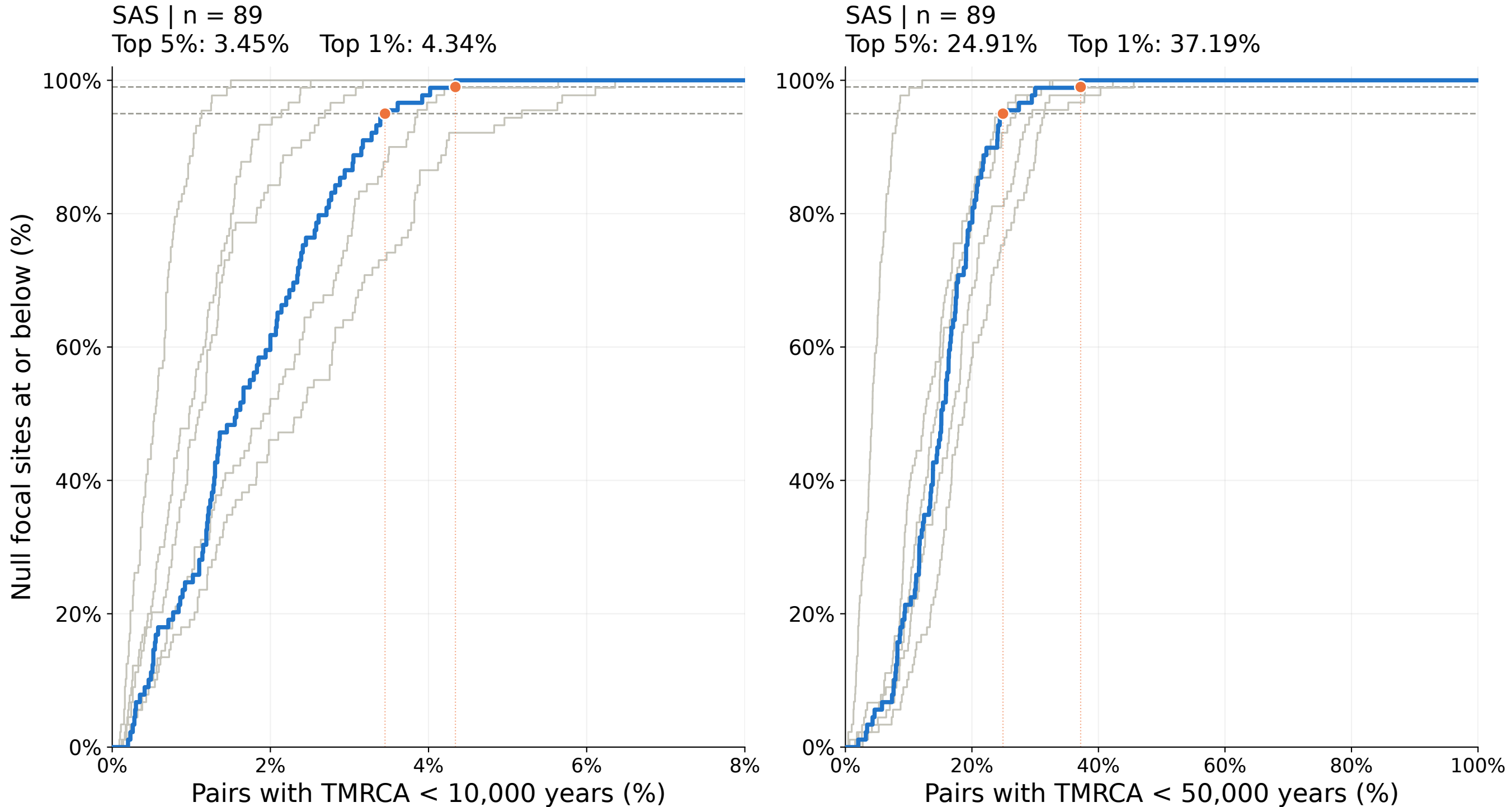
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Decoded TMRCA | All sampled pairs



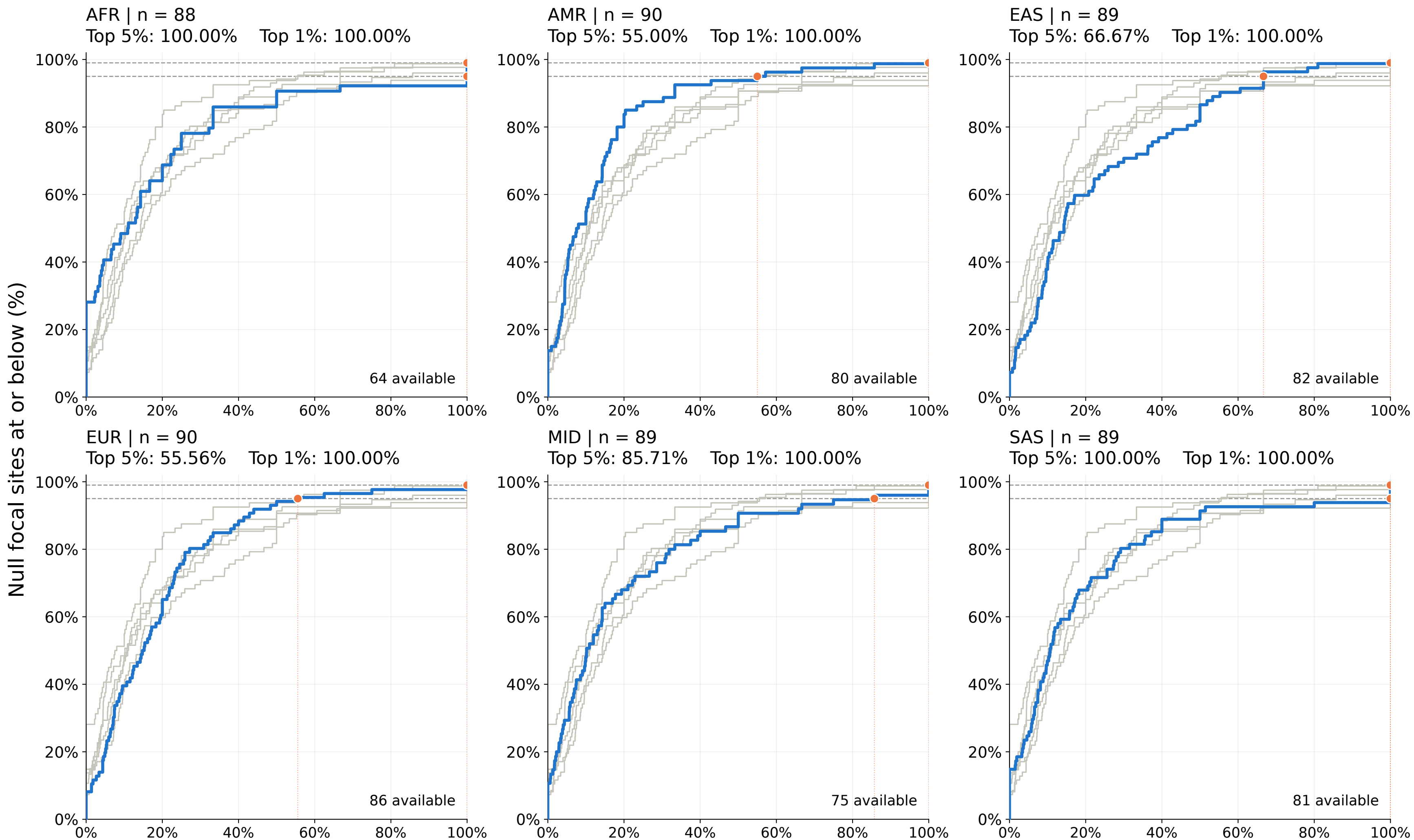
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Decoded TMRCA | All sampled pairs



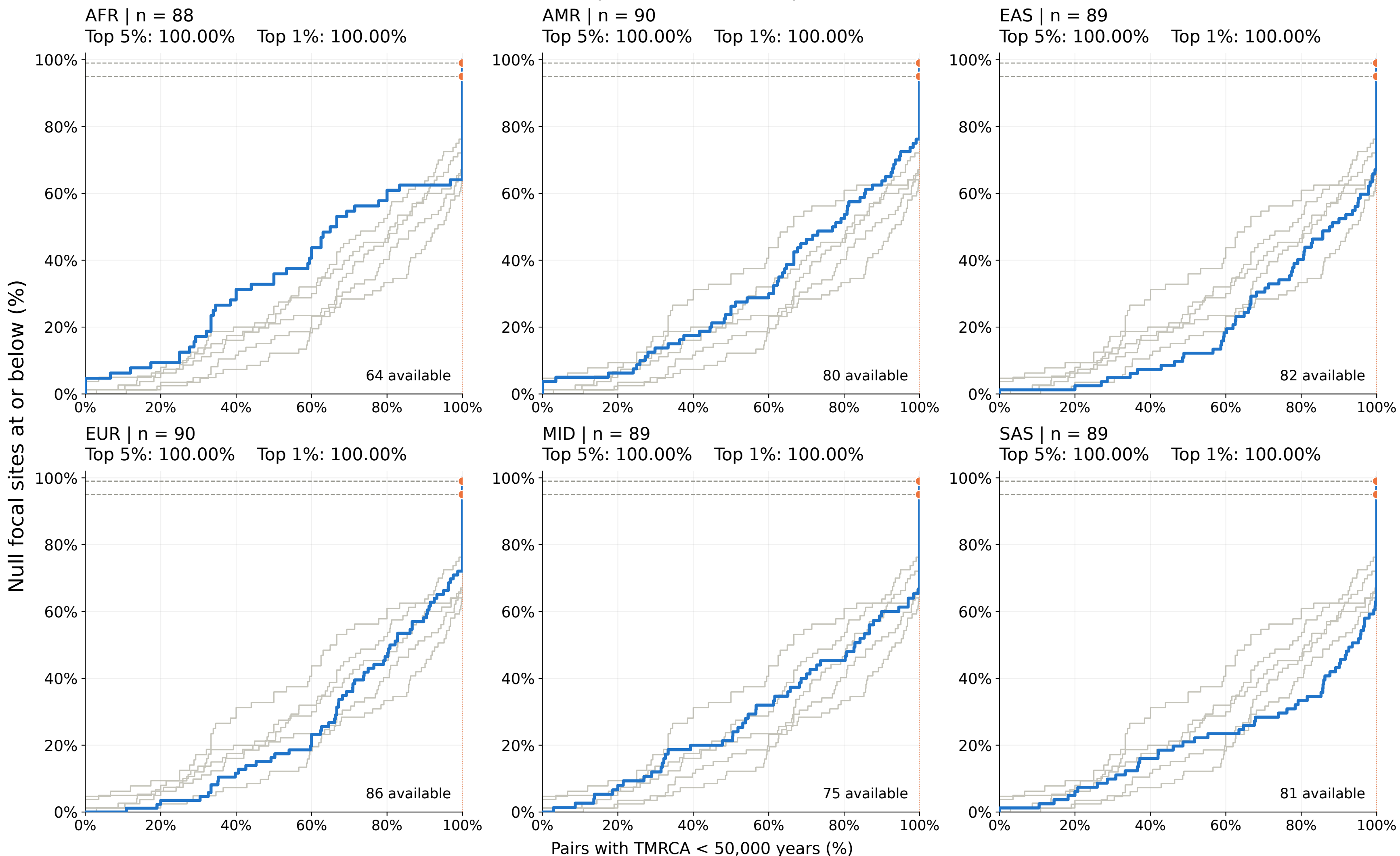
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | ALT/ALT pairs | T < 10,000 years



PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | ALT/ALT pairs | T < 50,000 years

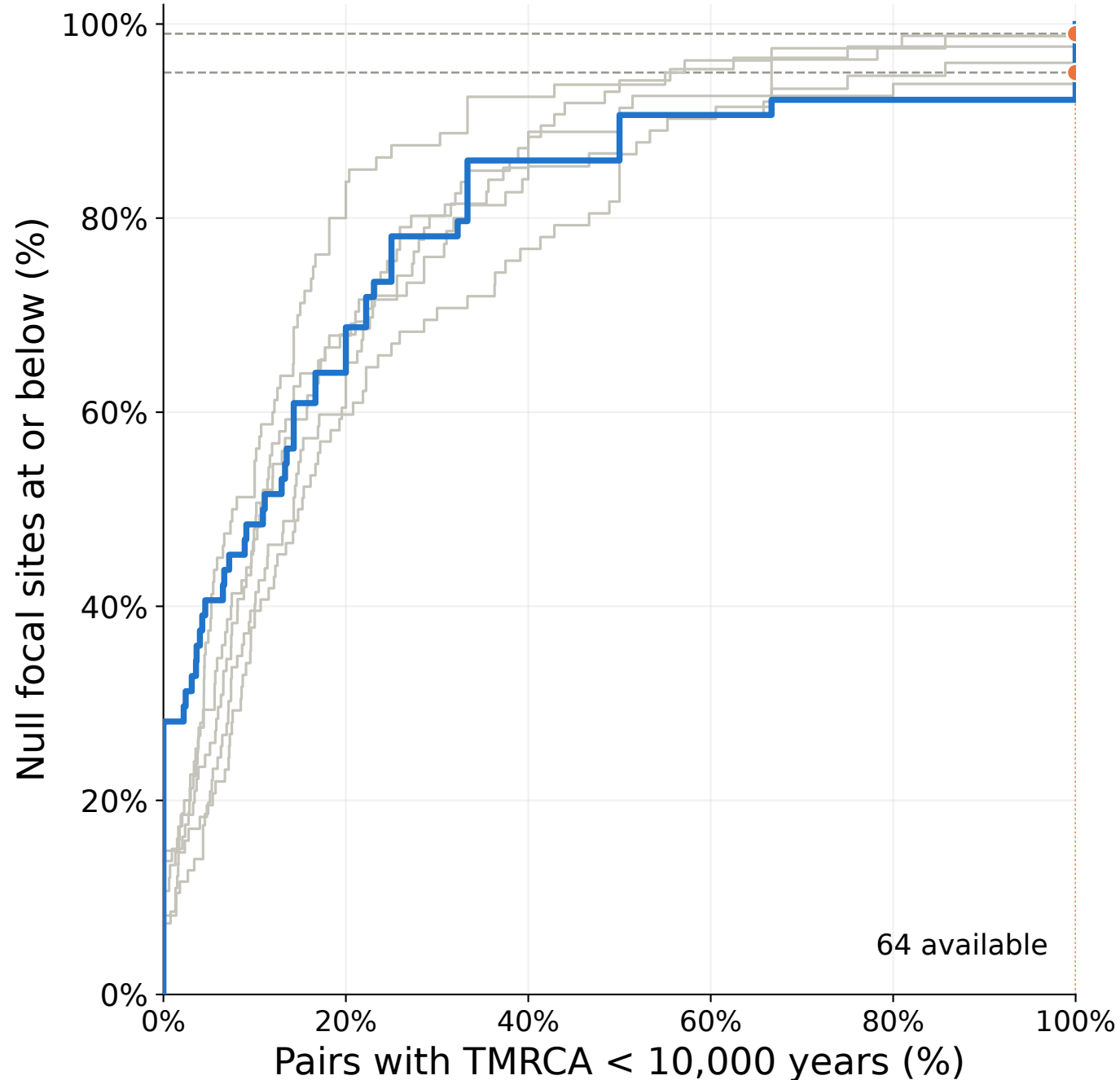


PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | ALT/ALT pairs

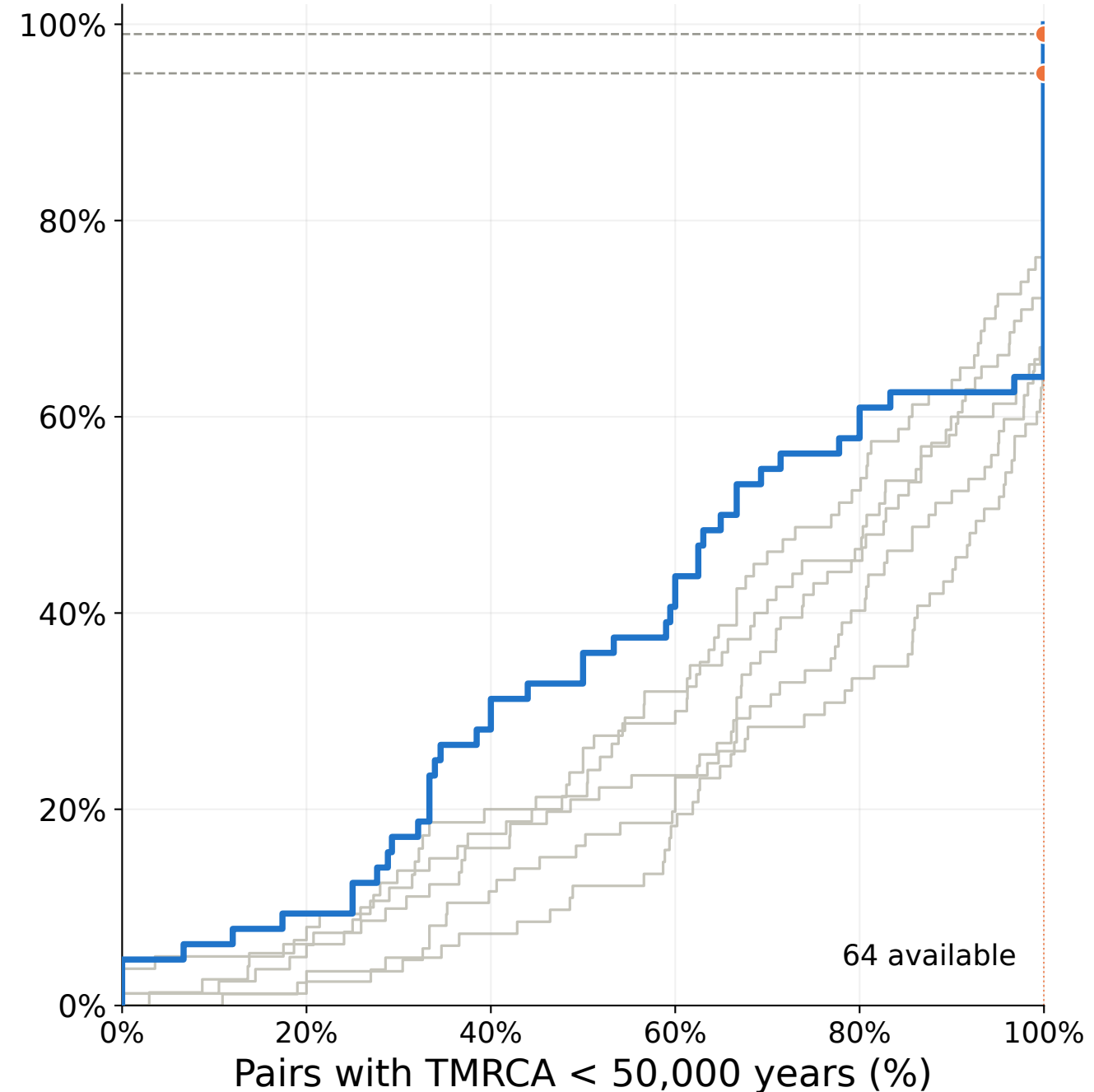
AFR | n = 88

Top 5%: 100.00% Top 1%: 100.00%



AFR | n = 88

Top 5%: 100.00% Top 1%: 100.00%



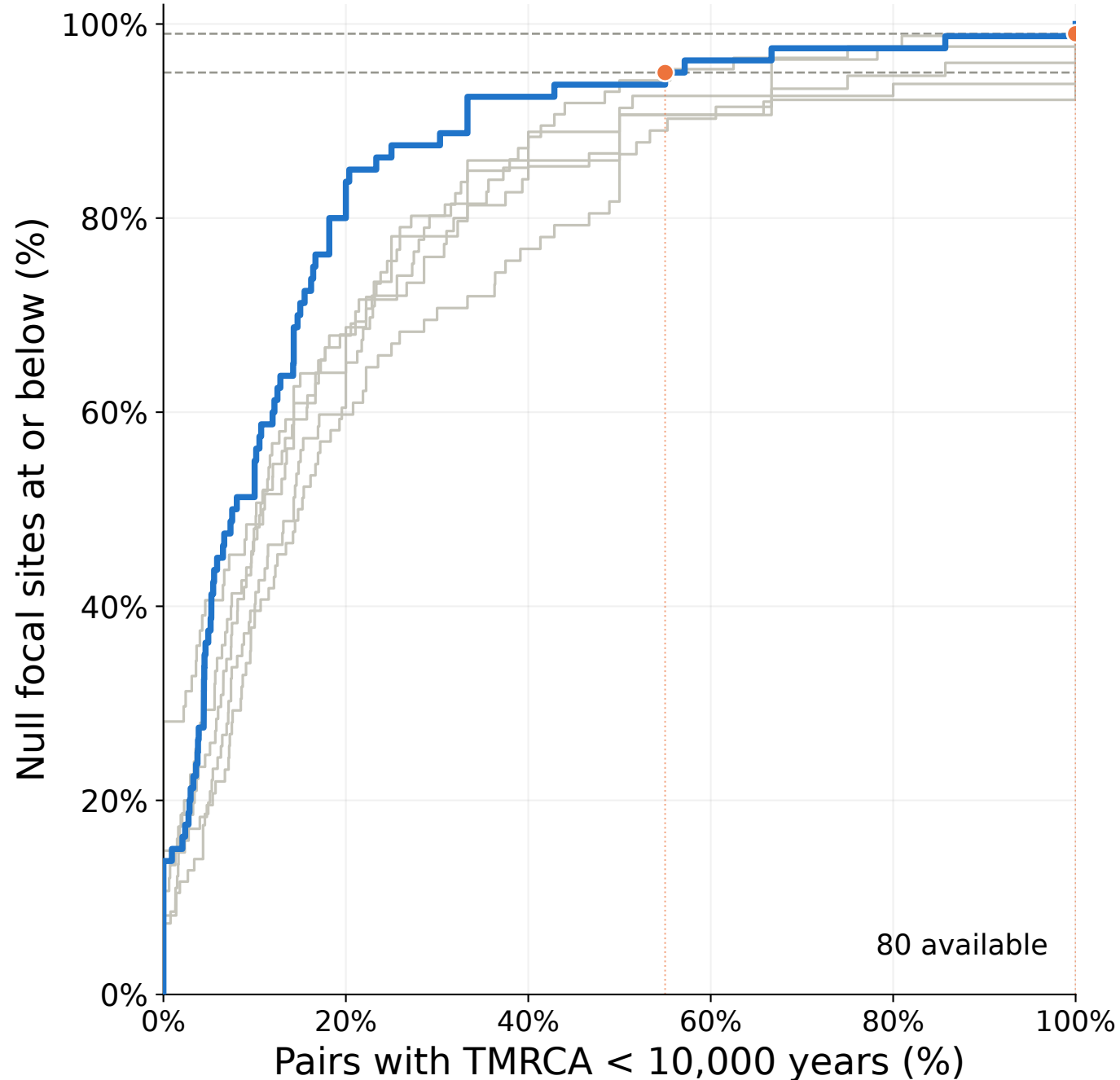
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | ALT/ALT pairs

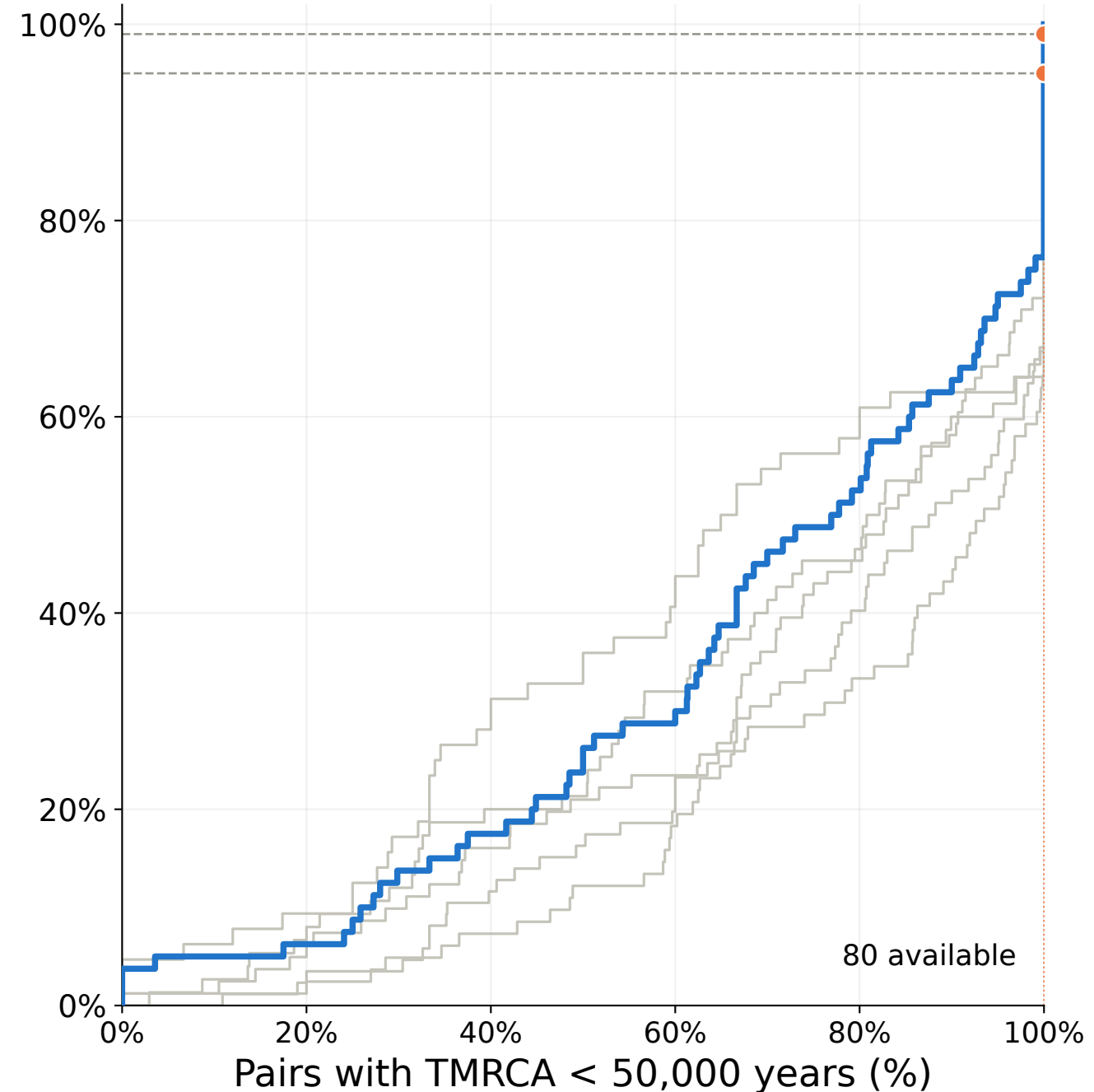
AMR | n = 90

Top 5%: 55.00% Top 1%: 100.00%



AMR | n = 90

Top 5%: 100.00% Top 1%: 100.00%



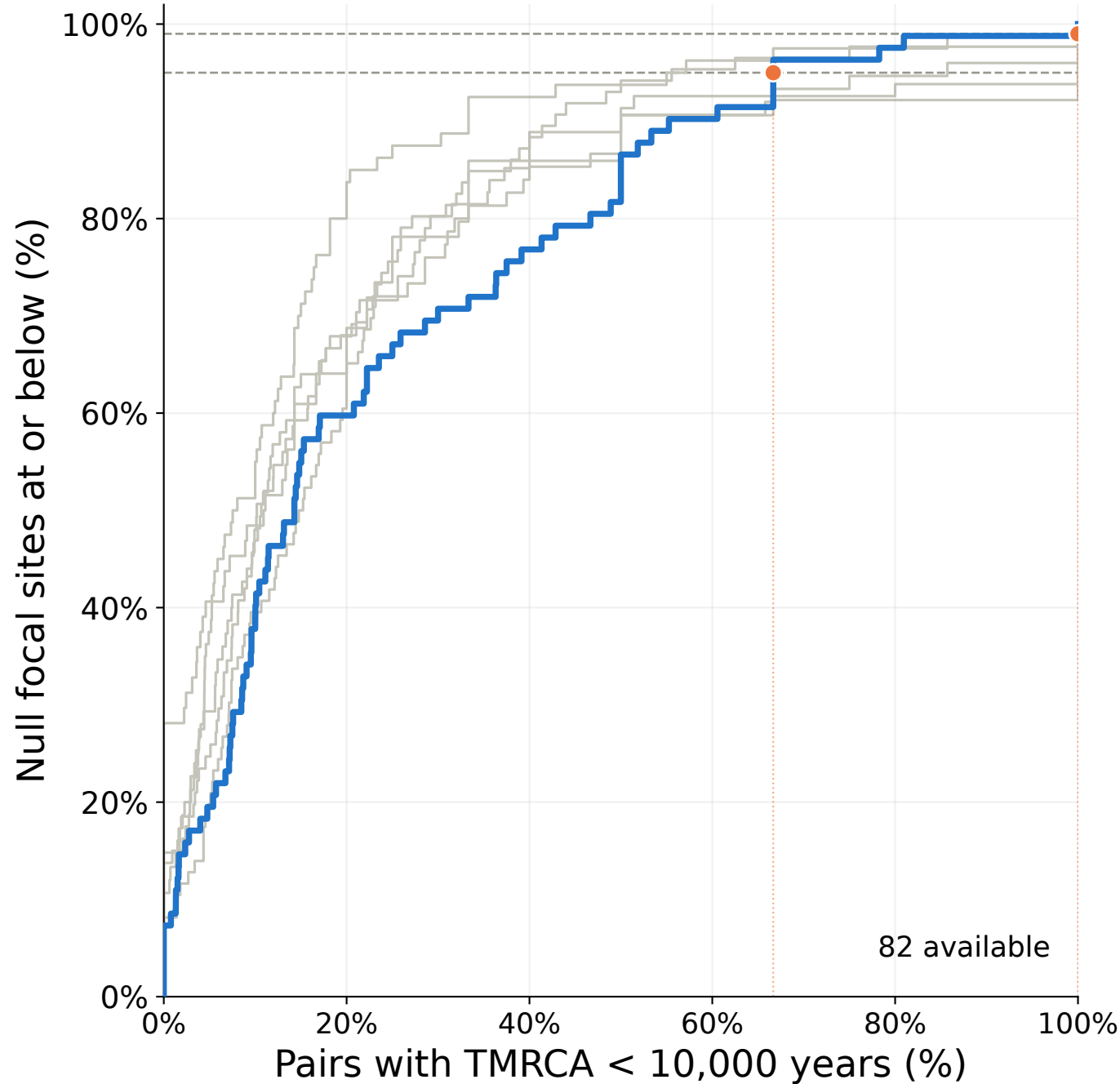
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

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Decoded TMRCA | ALT/ALT pairs

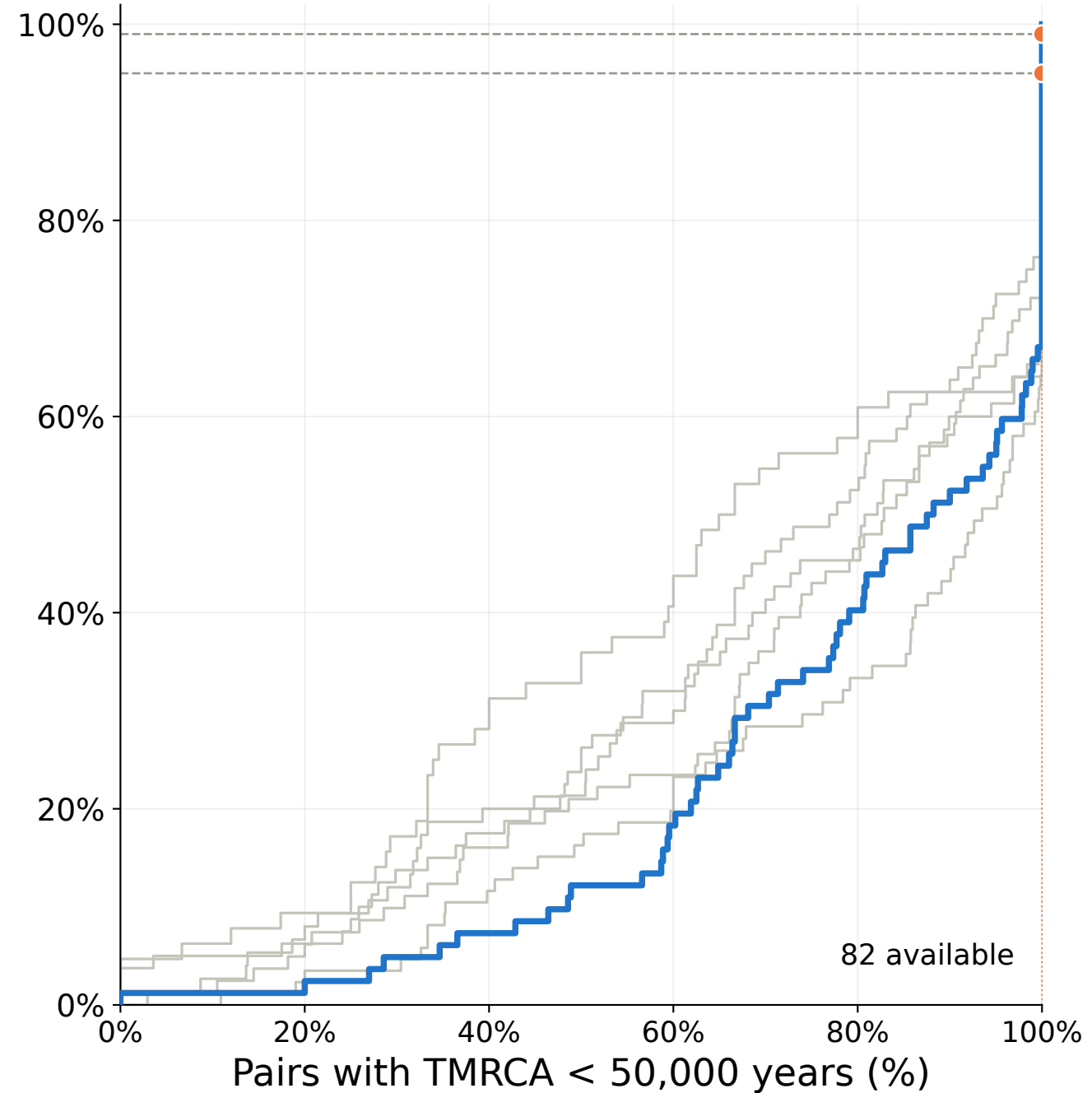
EAS | n = 89

Top 5%: 66.67% Top 1%: 100.00%



EAS | n = 89

Top 5%: 100.00% Top 1%: 100.00%



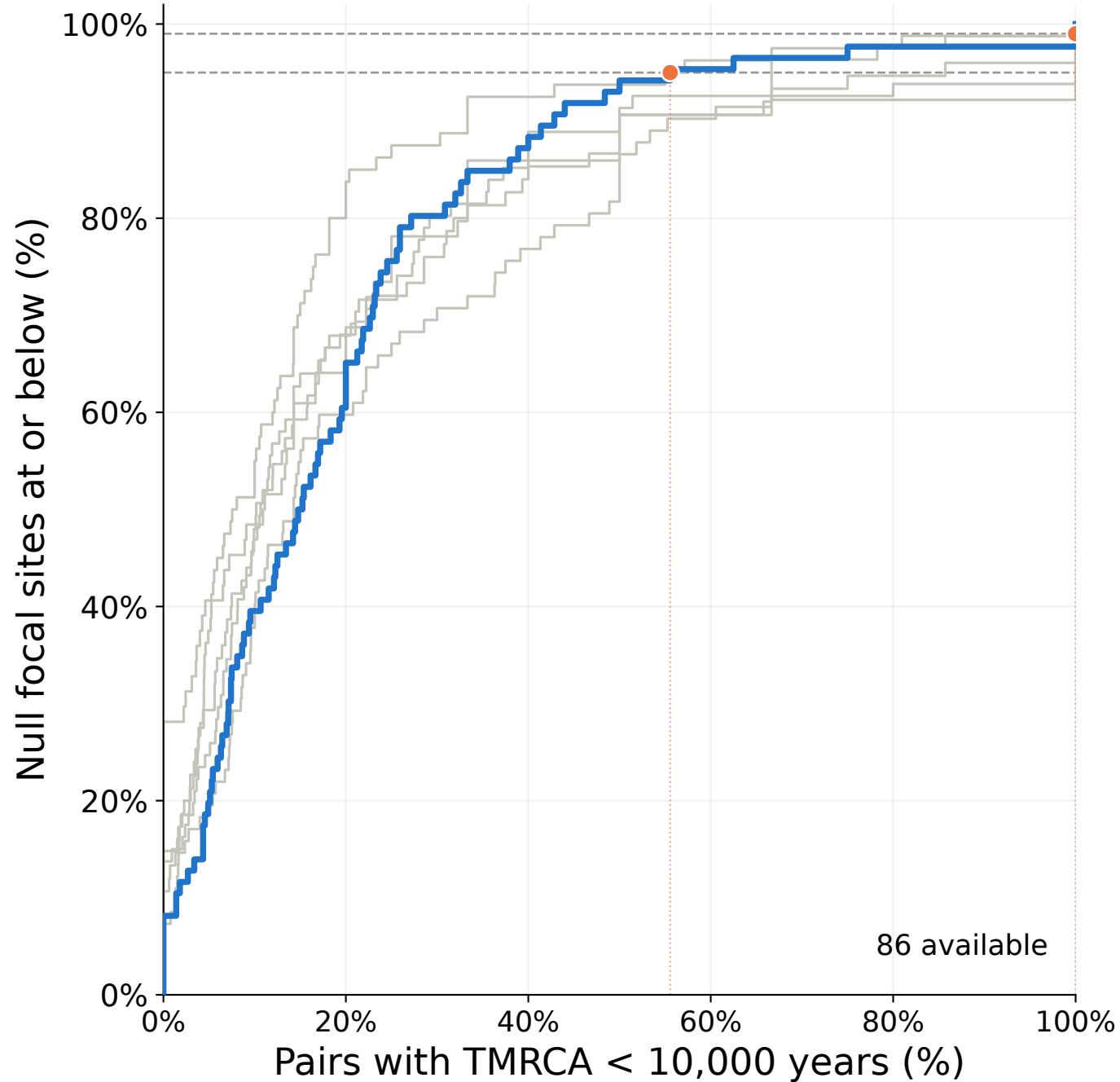
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

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Decoded TMRCA | ALT/ALT pairs

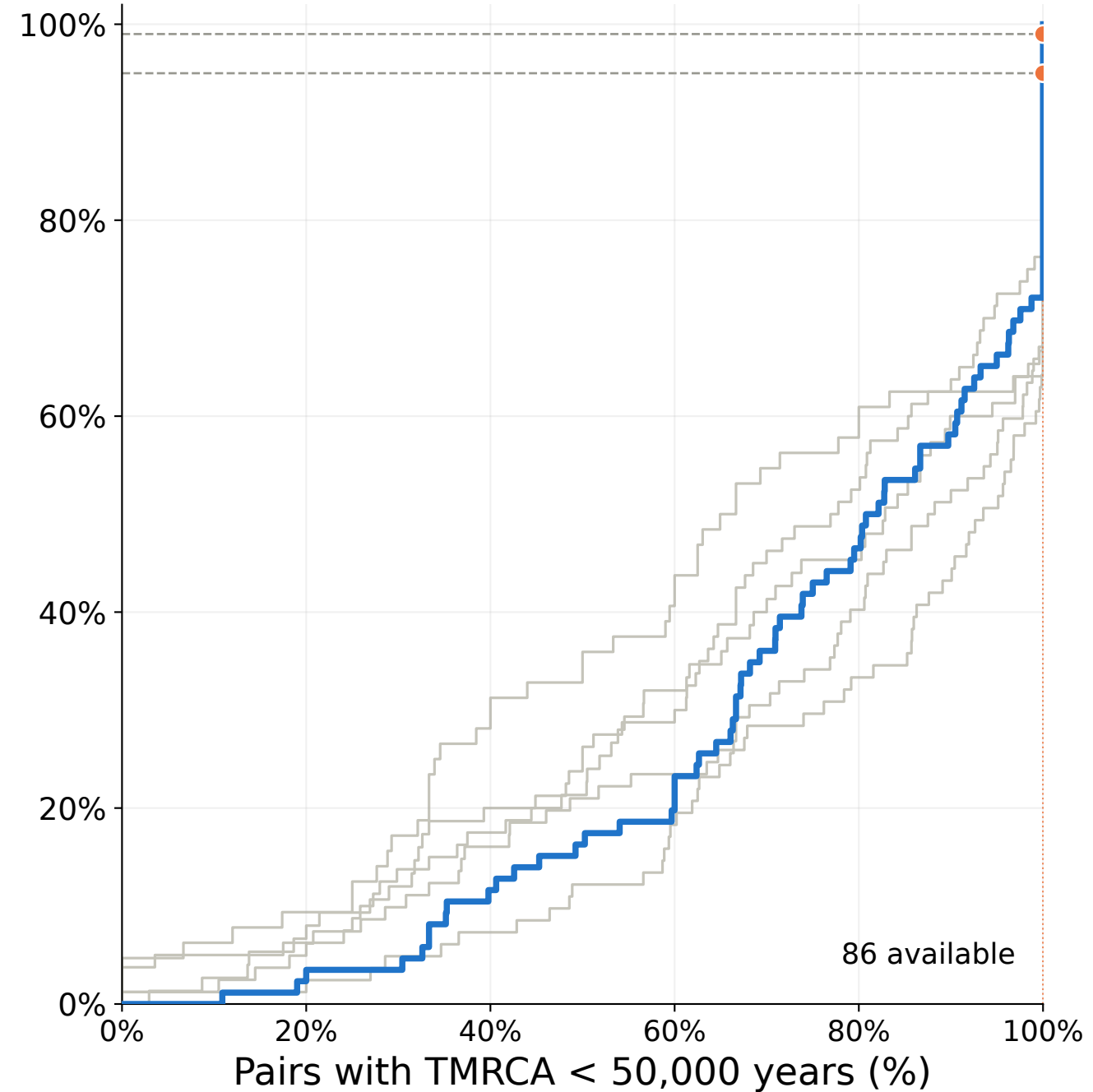
EUR | n = 90

Top 5%: 55.56% Top 1%: 100.00%



EUR | n = 90

Top 5%: 100.00% Top 1%: 100.00%



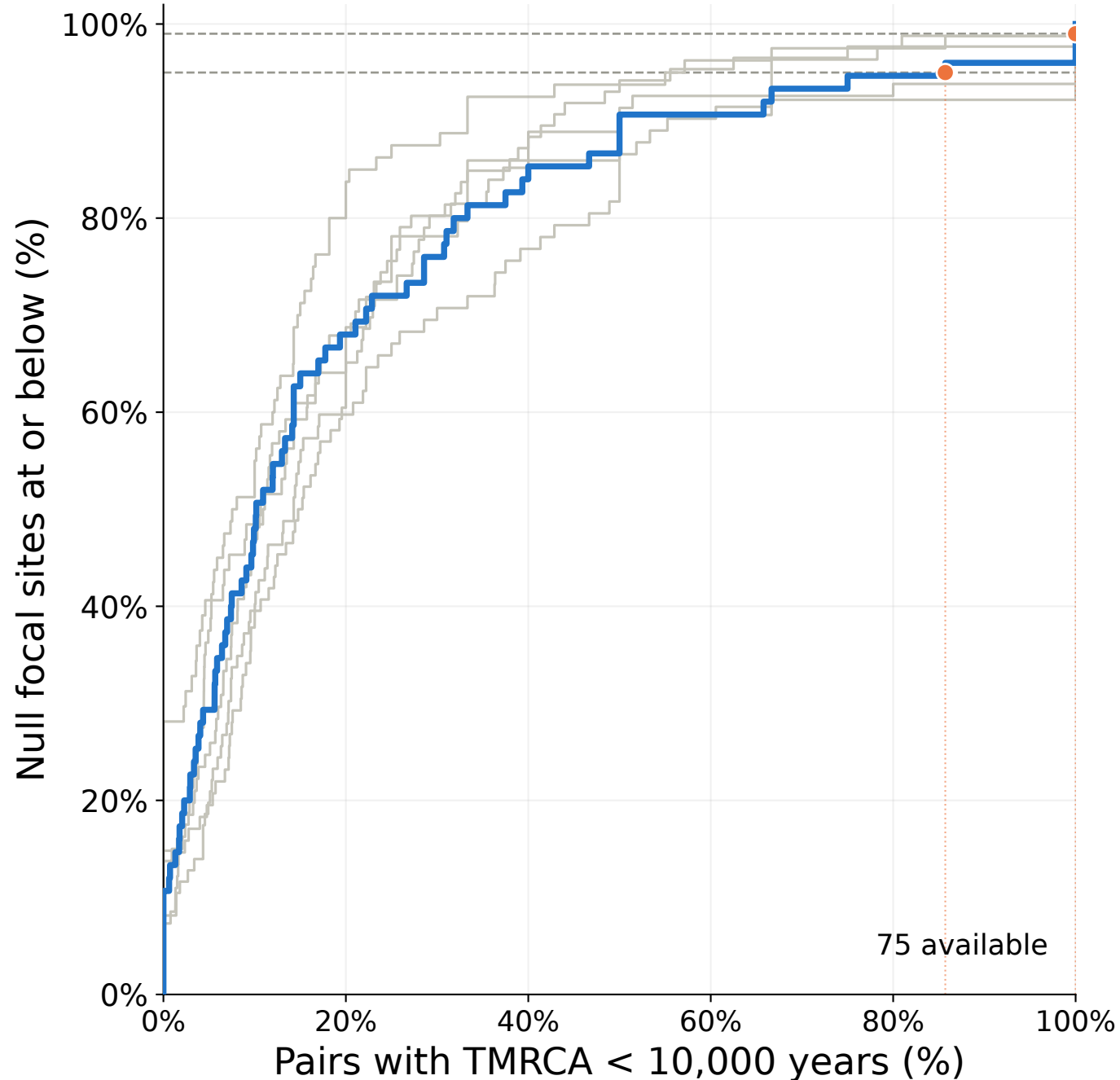
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

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Decoded TMRCA | ALT/ALT pairs

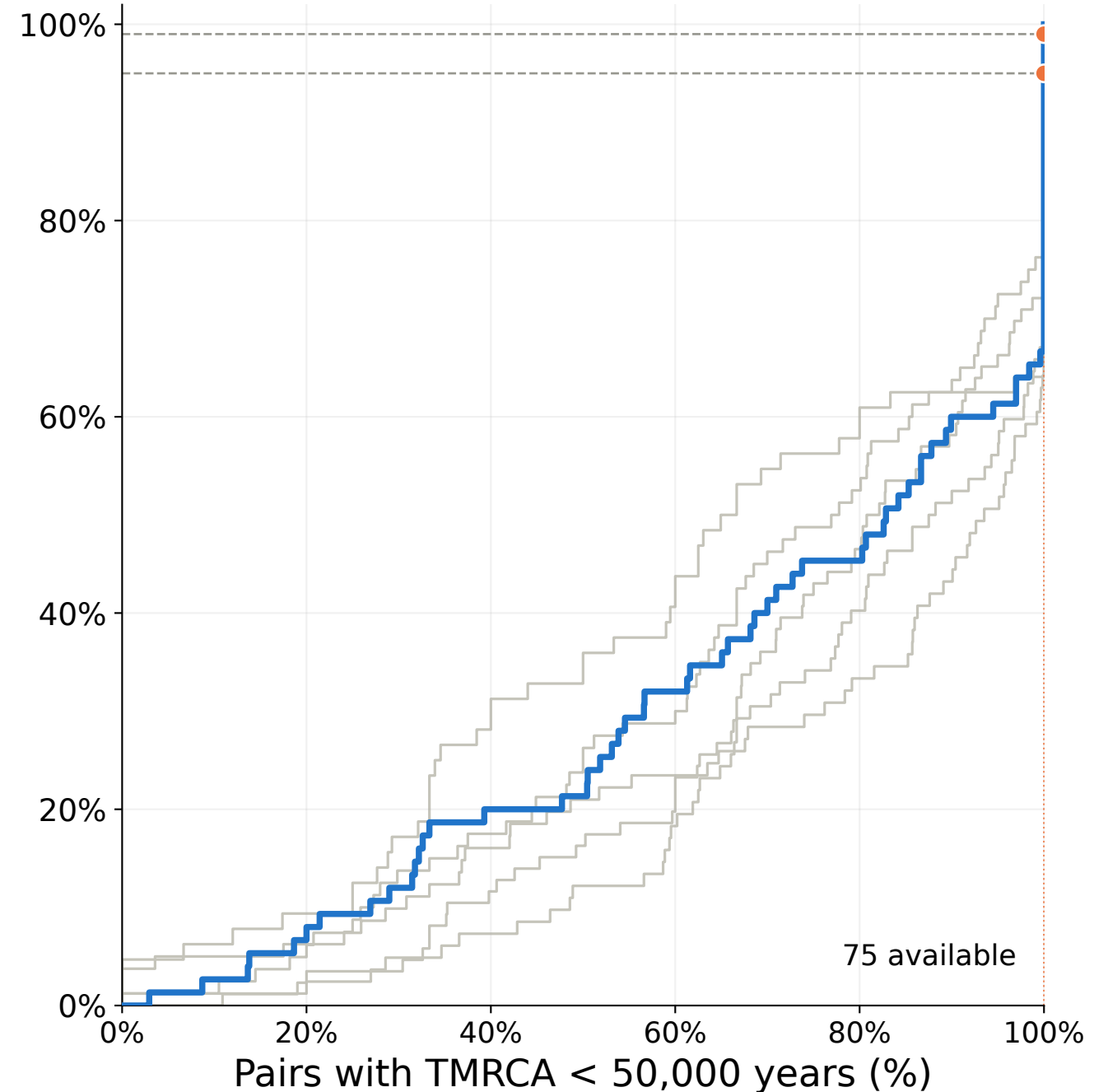
MID | n = 89

Top 5%: 85.71% Top 1%: 100.00%



MID | n = 89

Top 5%: 100.00% Top 1%: 100.00%

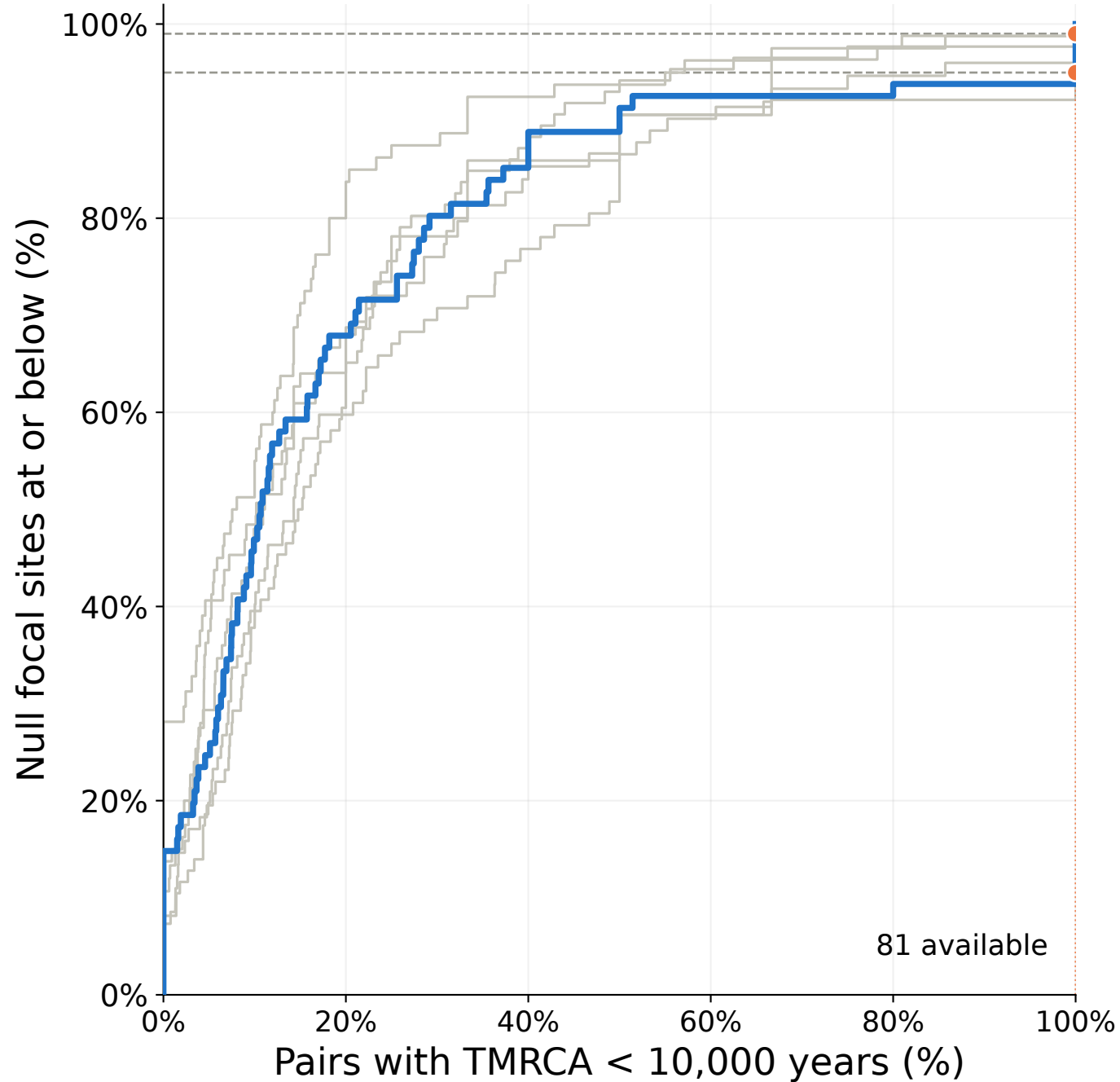


PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Decoded TMRCA | ALT/ALT pairs

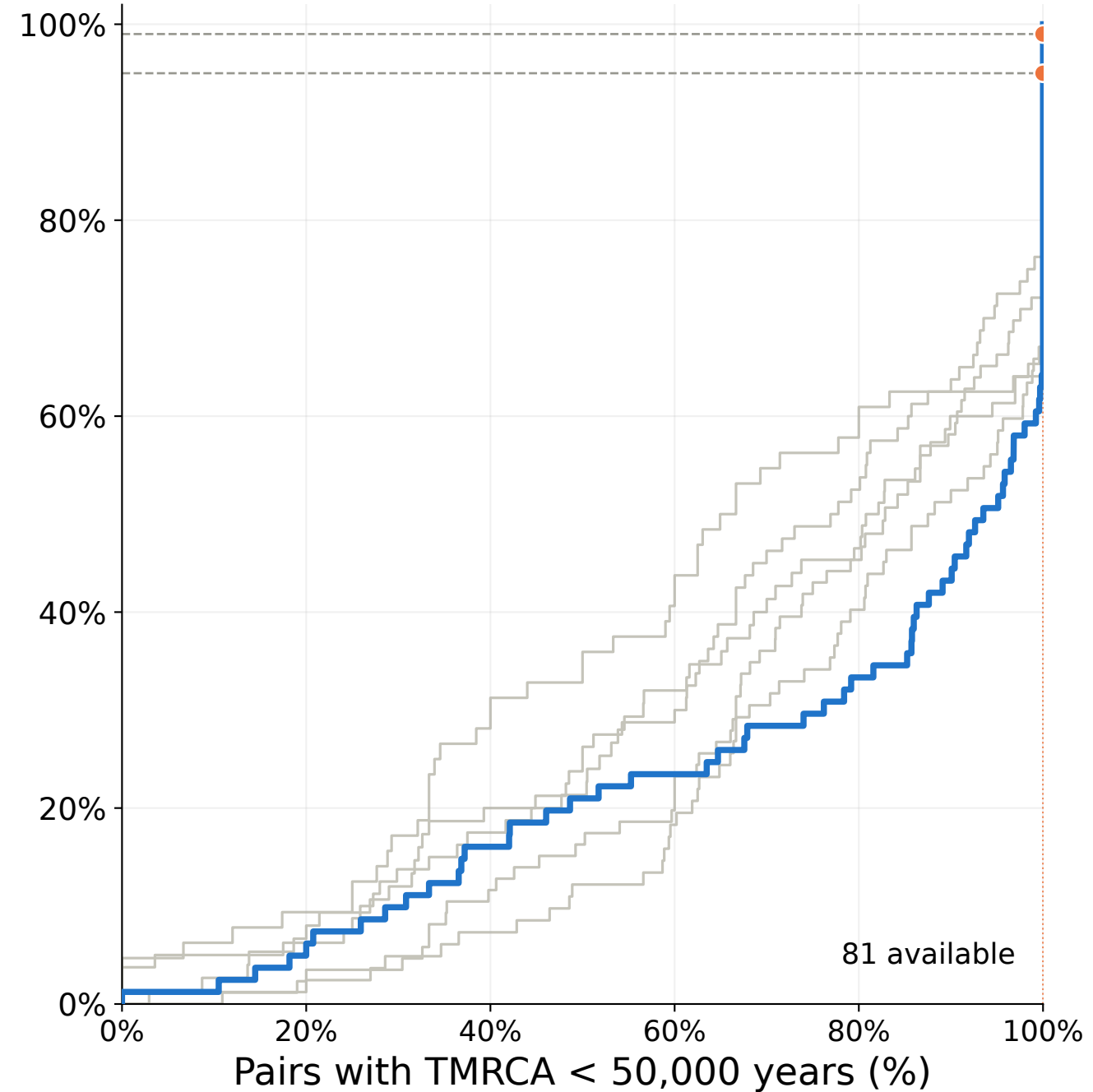
SAS | n = 89

Top 5%: 100.00% Top 1%: 100.00%



SAS | n = 89

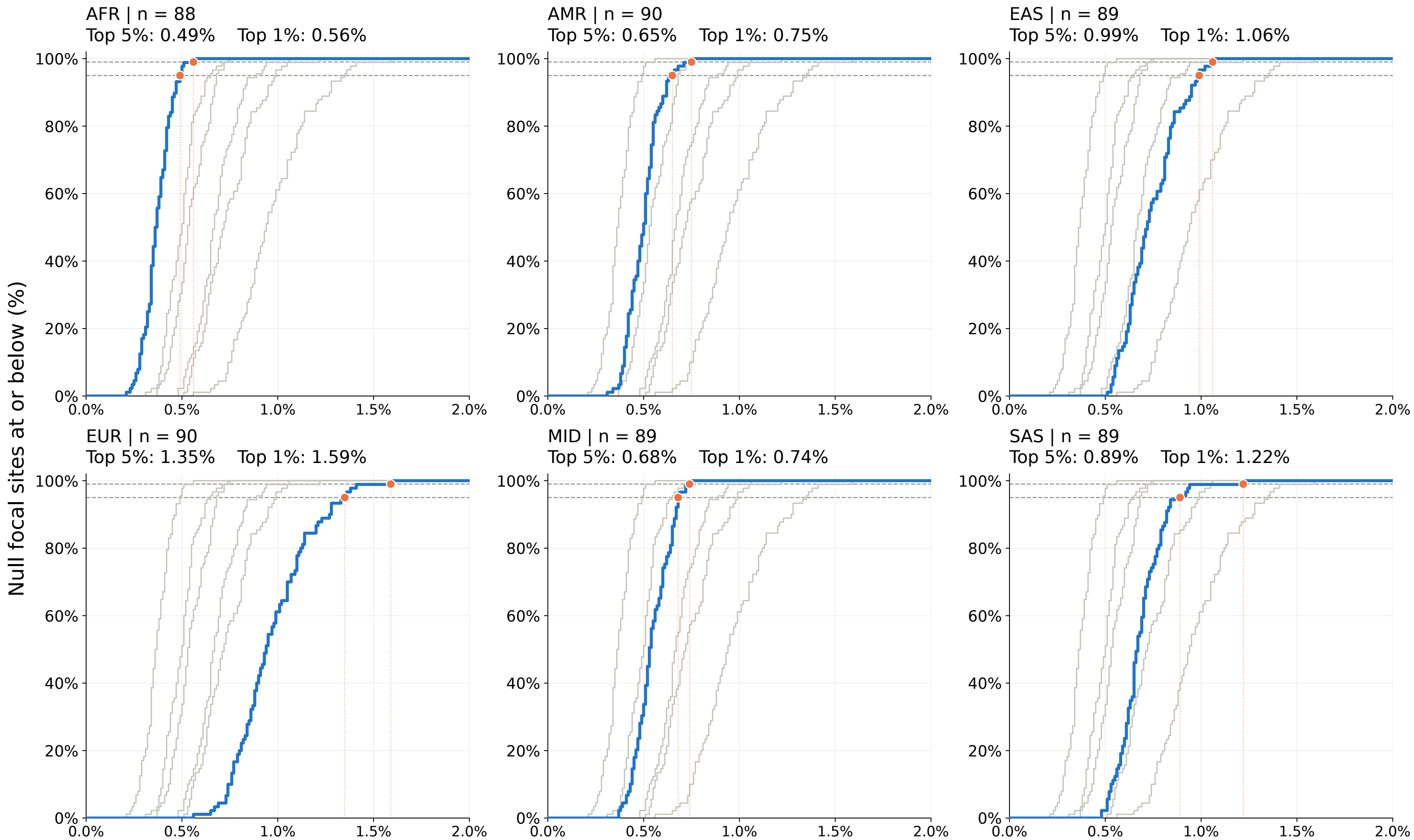
Top 5%: 100.00% Top 1%: 100.00%



PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | All sampled pairs | T < 10,000 years

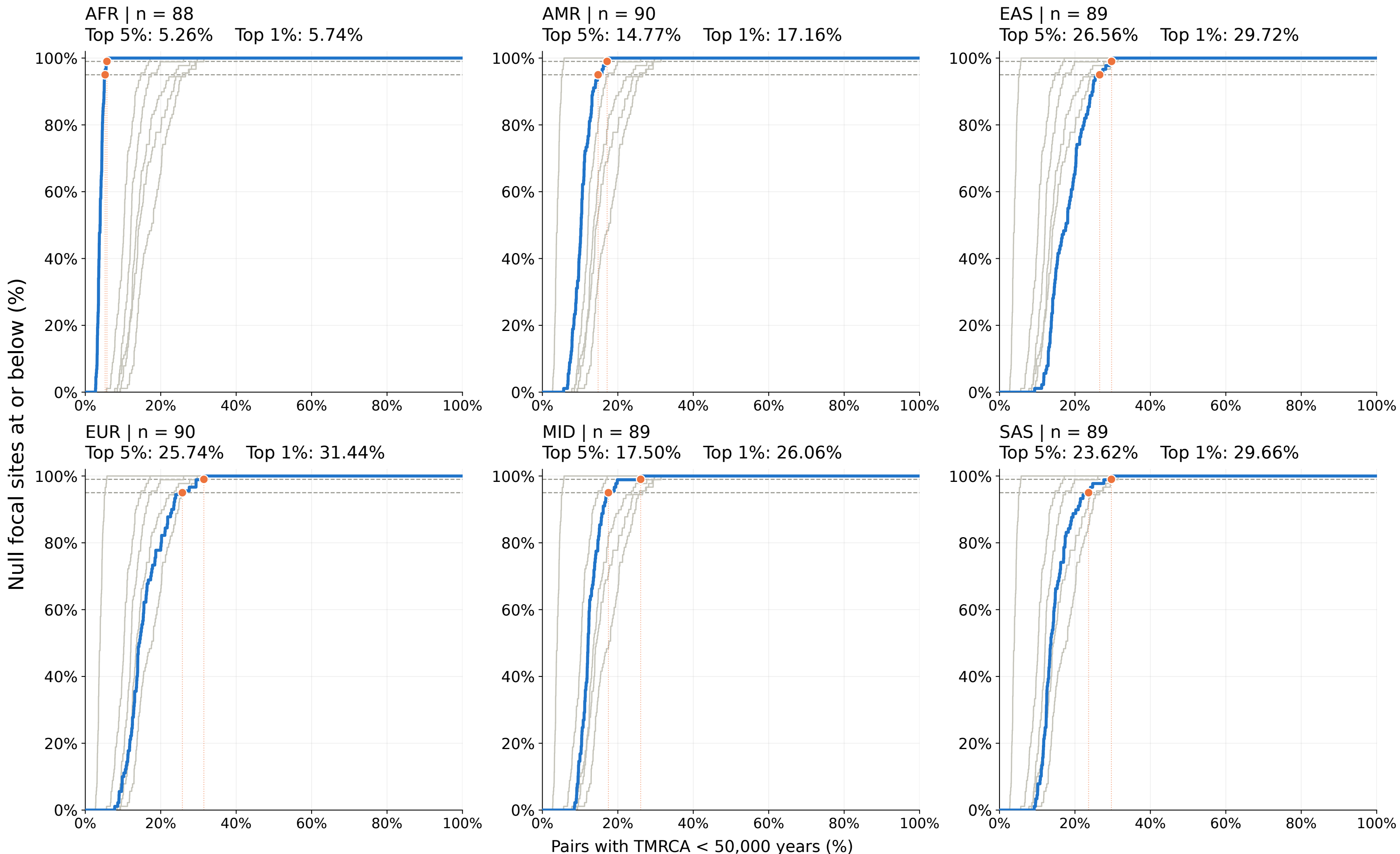


Pairs with TMRCA < 10,000 years (%)

PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

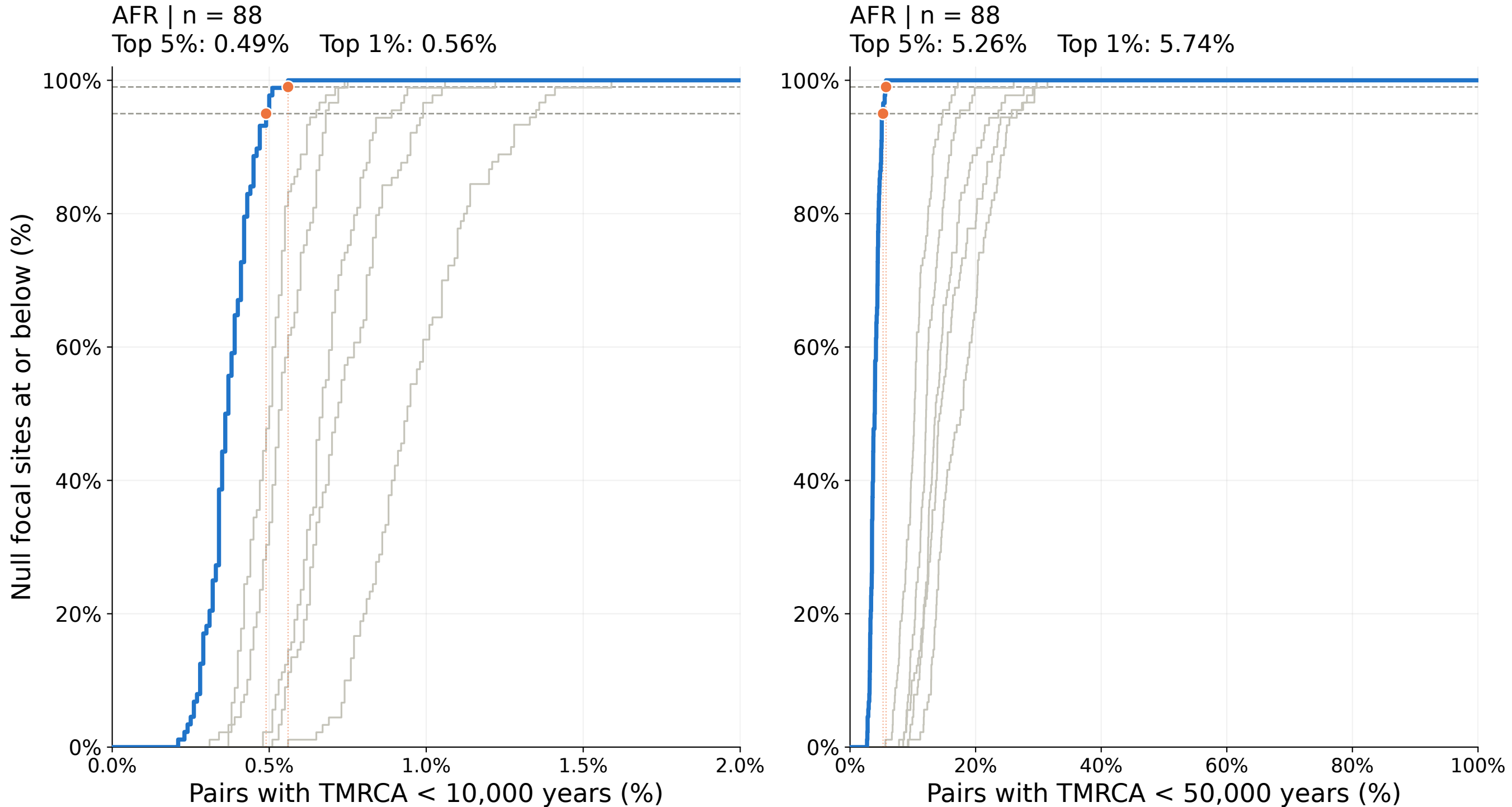
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | All sampled pairs | T < 50,000 years



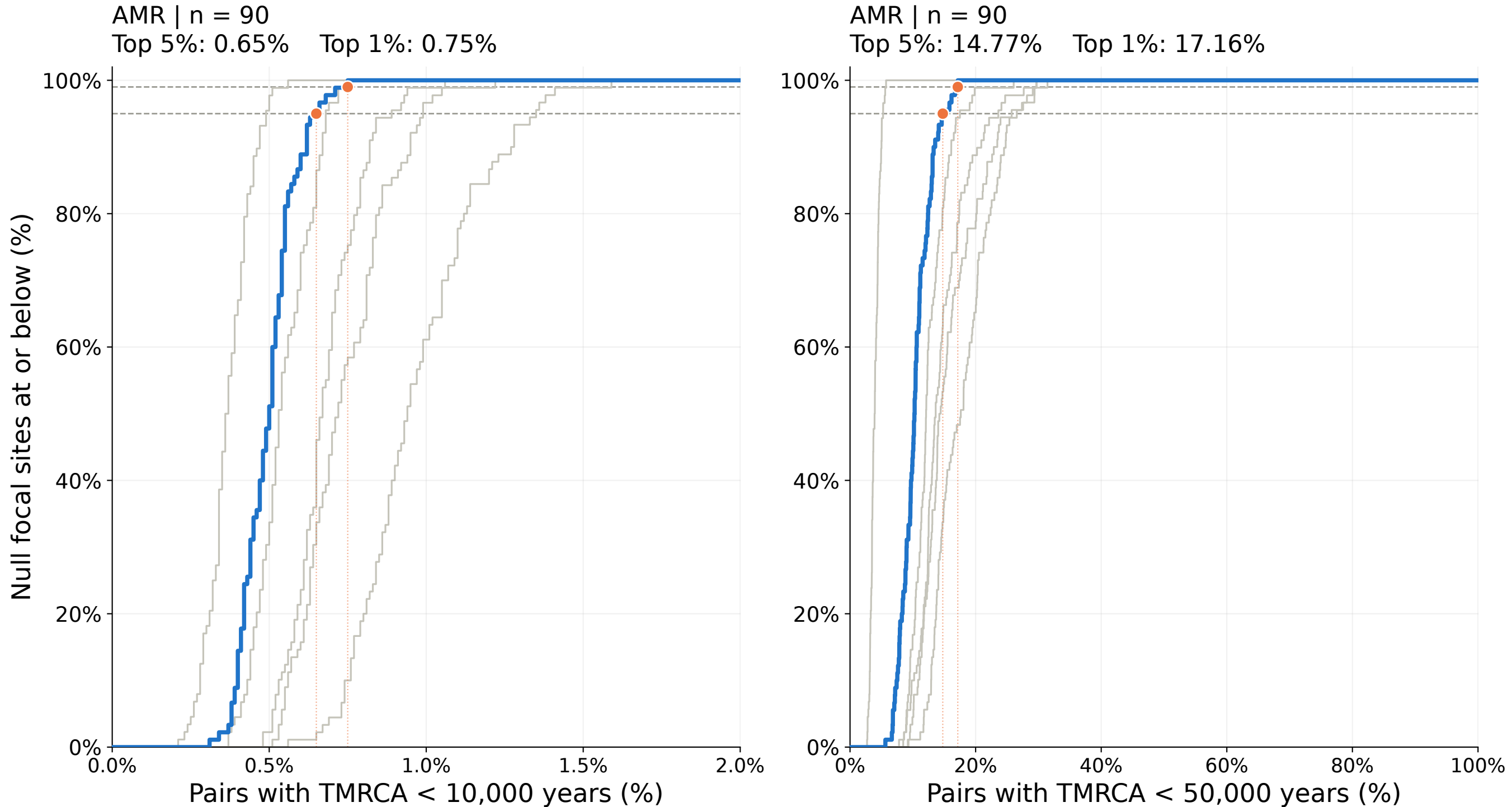
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | All sampled pairs



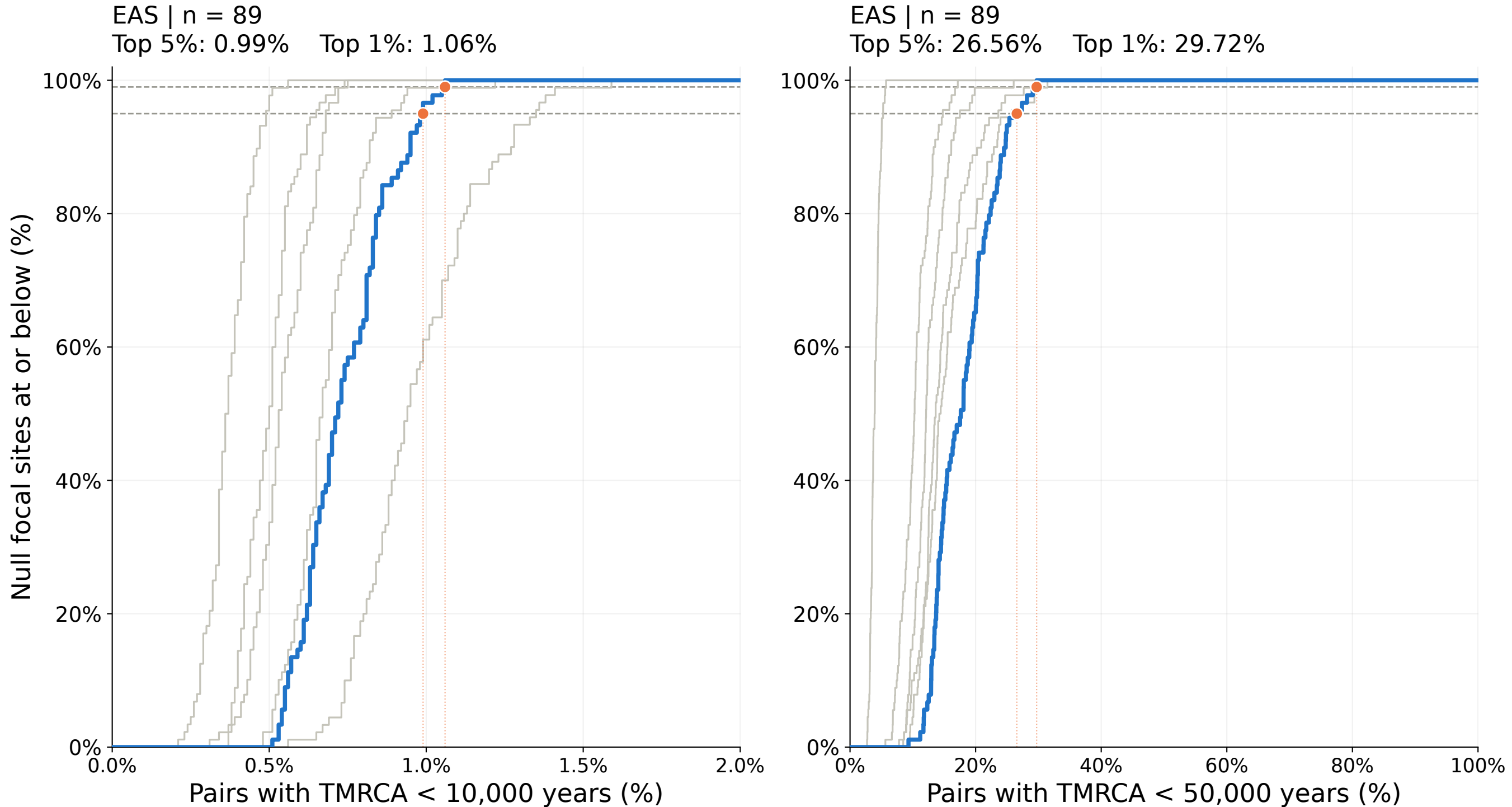
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | All sampled pairs



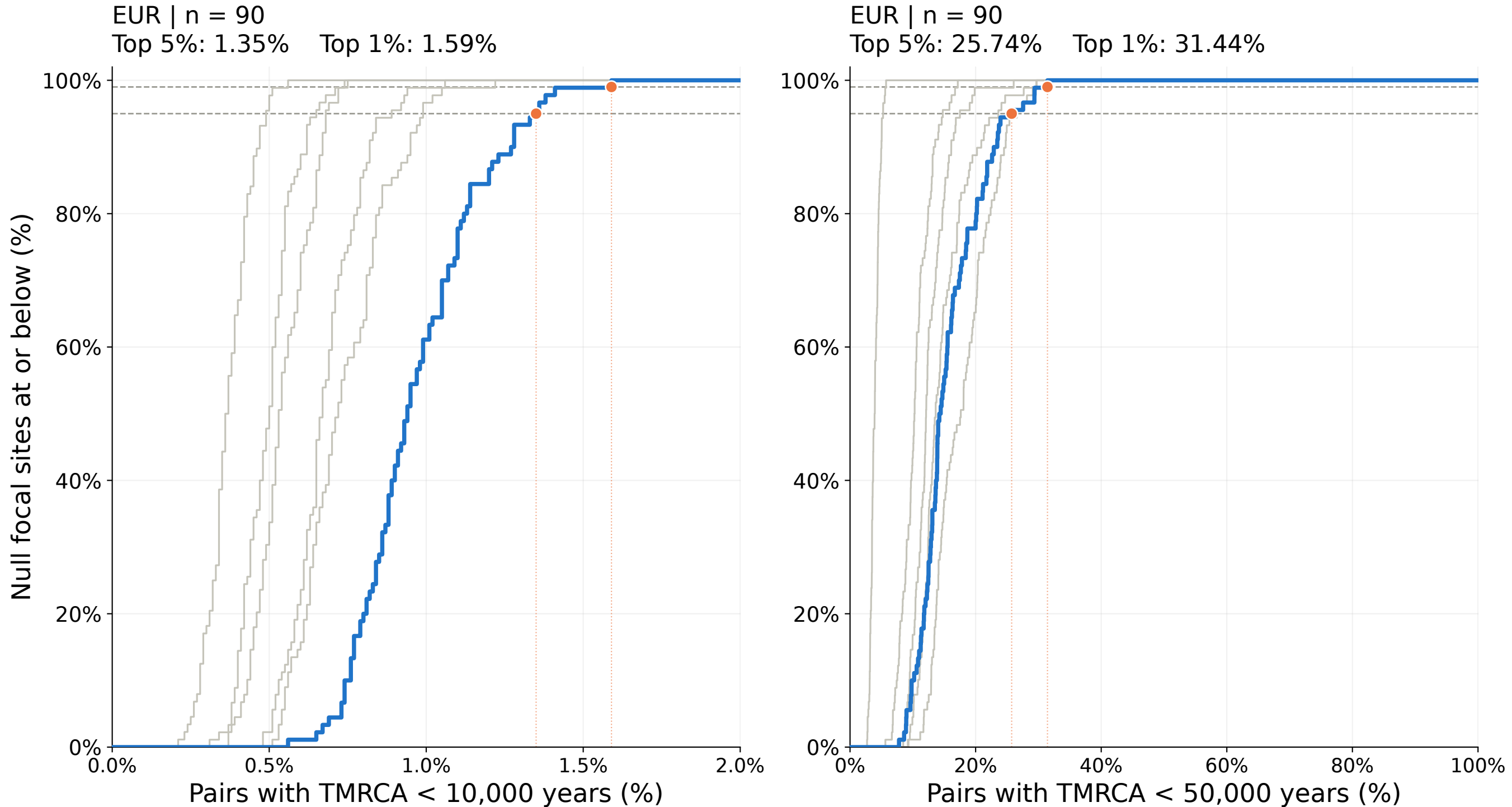
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
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Truth TMRCA | All sampled pairs



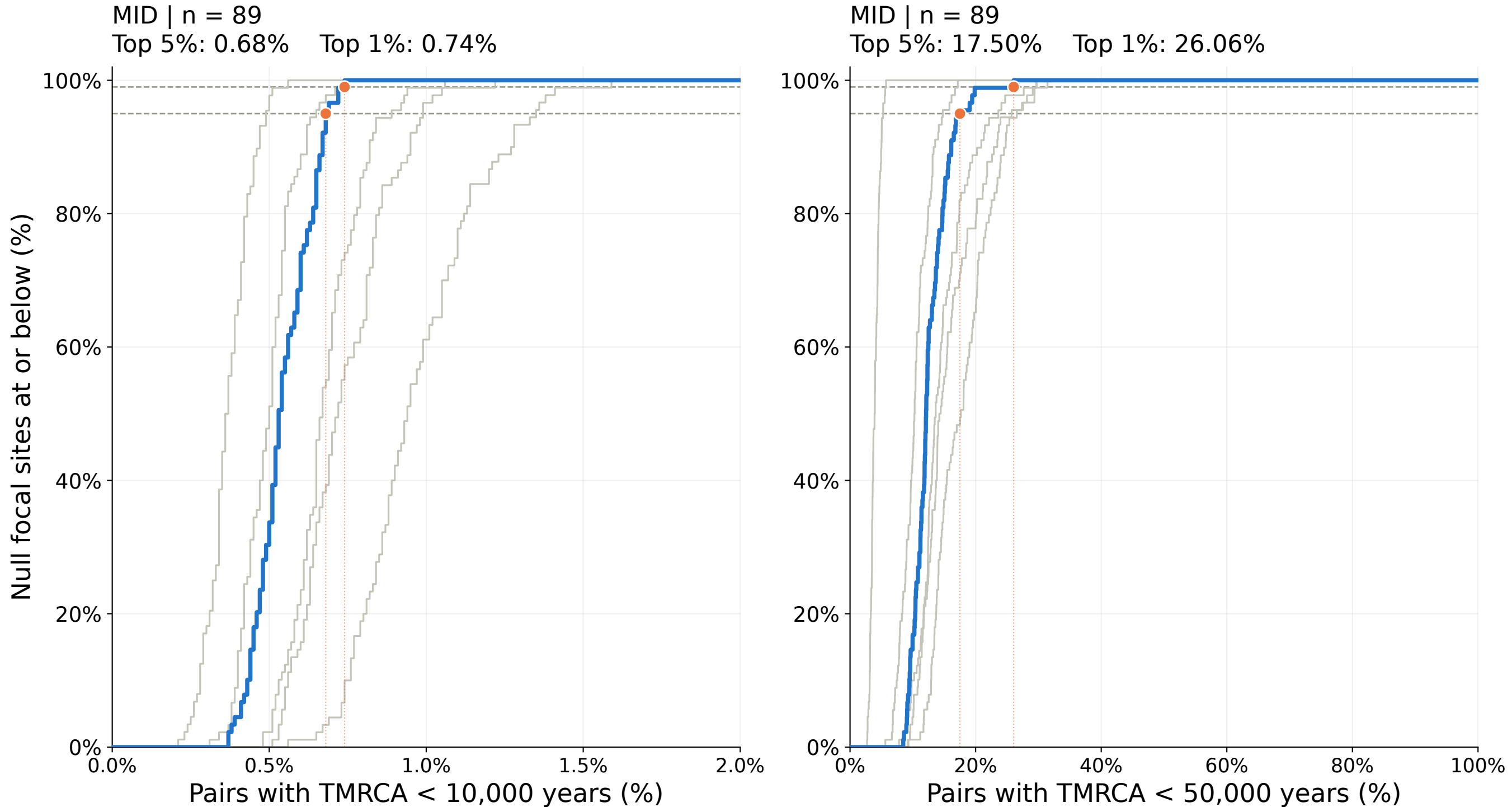
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | All sampled pairs



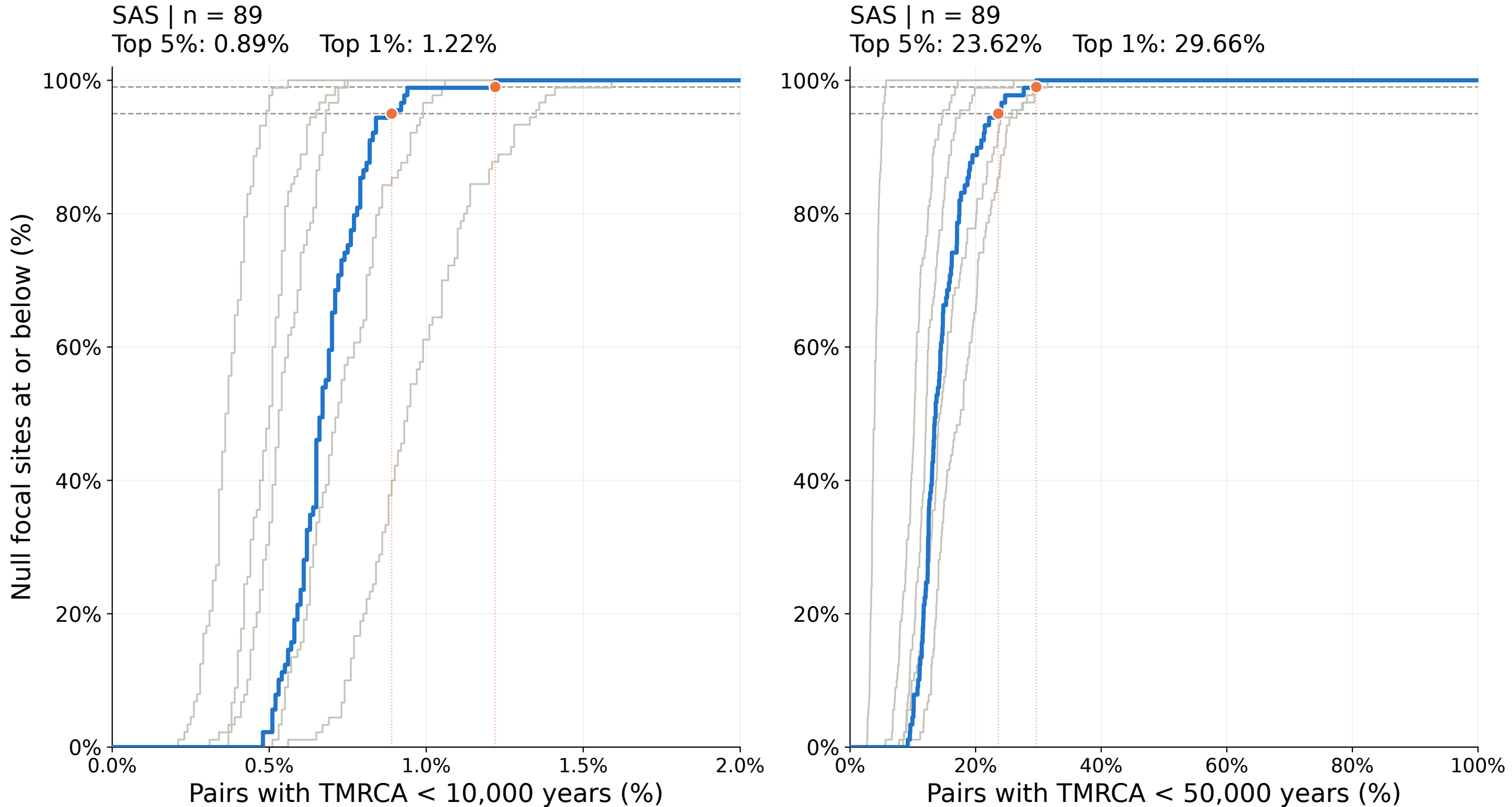
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Truth TMRCA | All sampled pairs



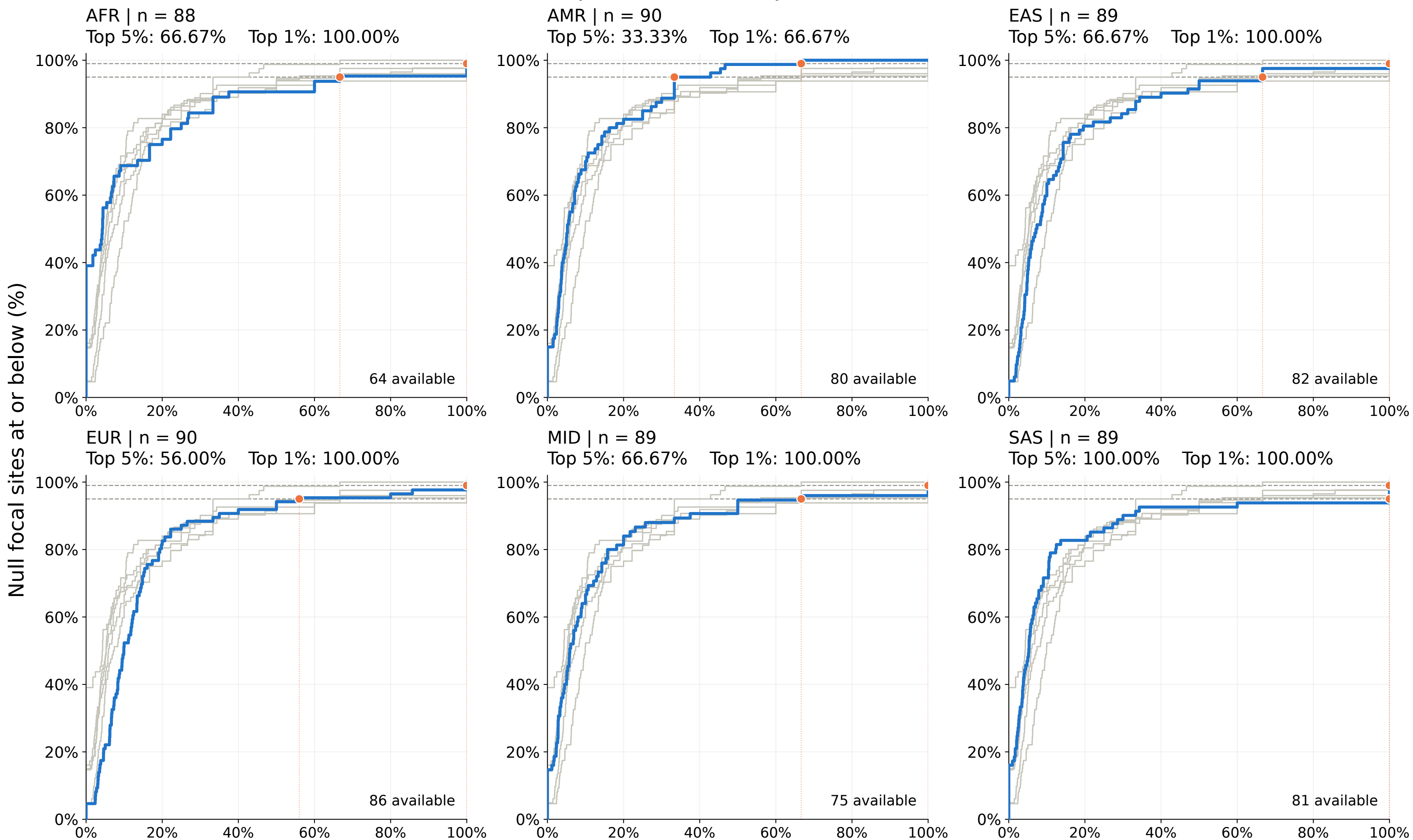
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Truth TMRCA | All sampled pairs



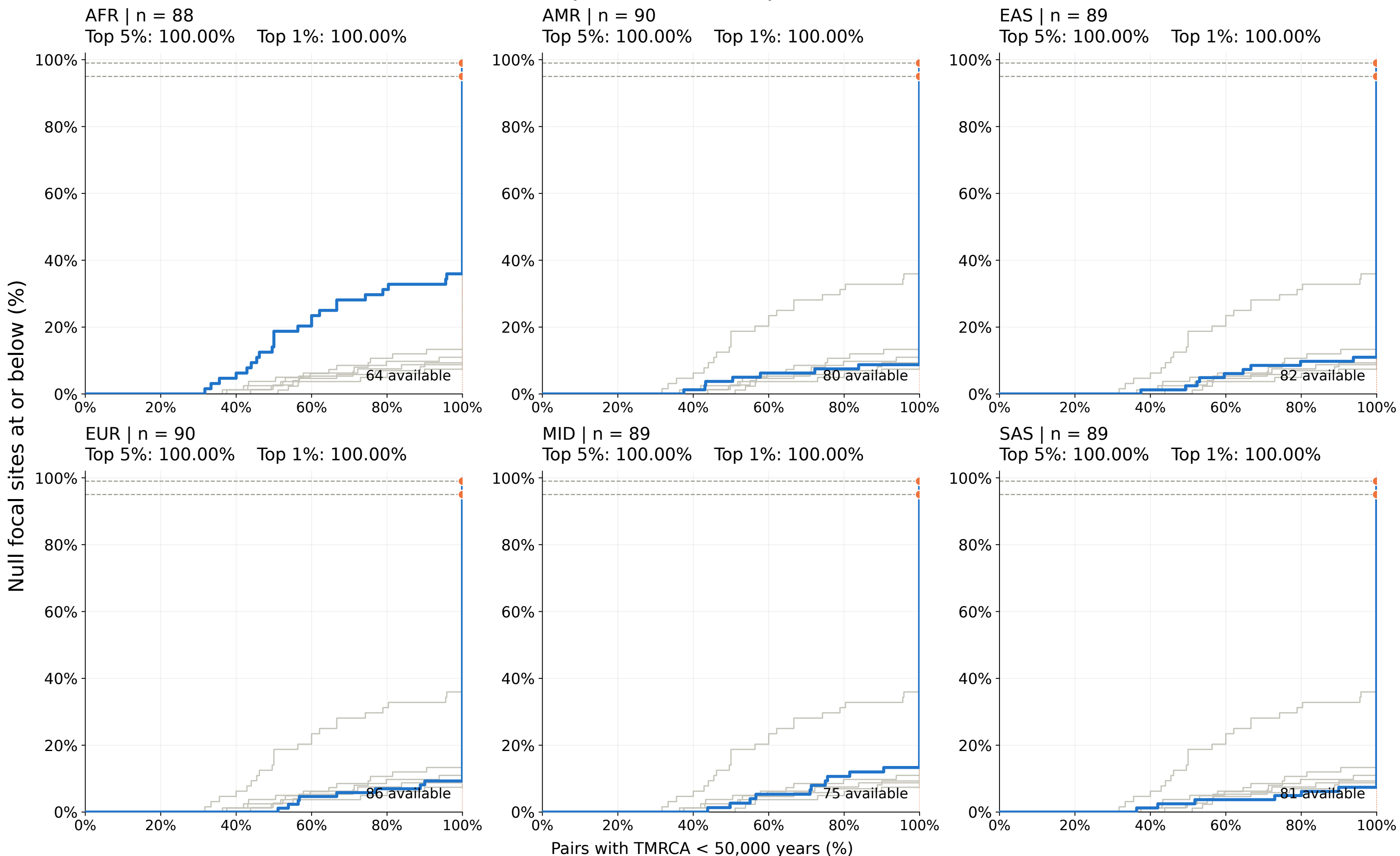
PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | ALT/ALT pairs | T < 10,000 years



PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | ALT/ALT pairs | T < 50,000 years

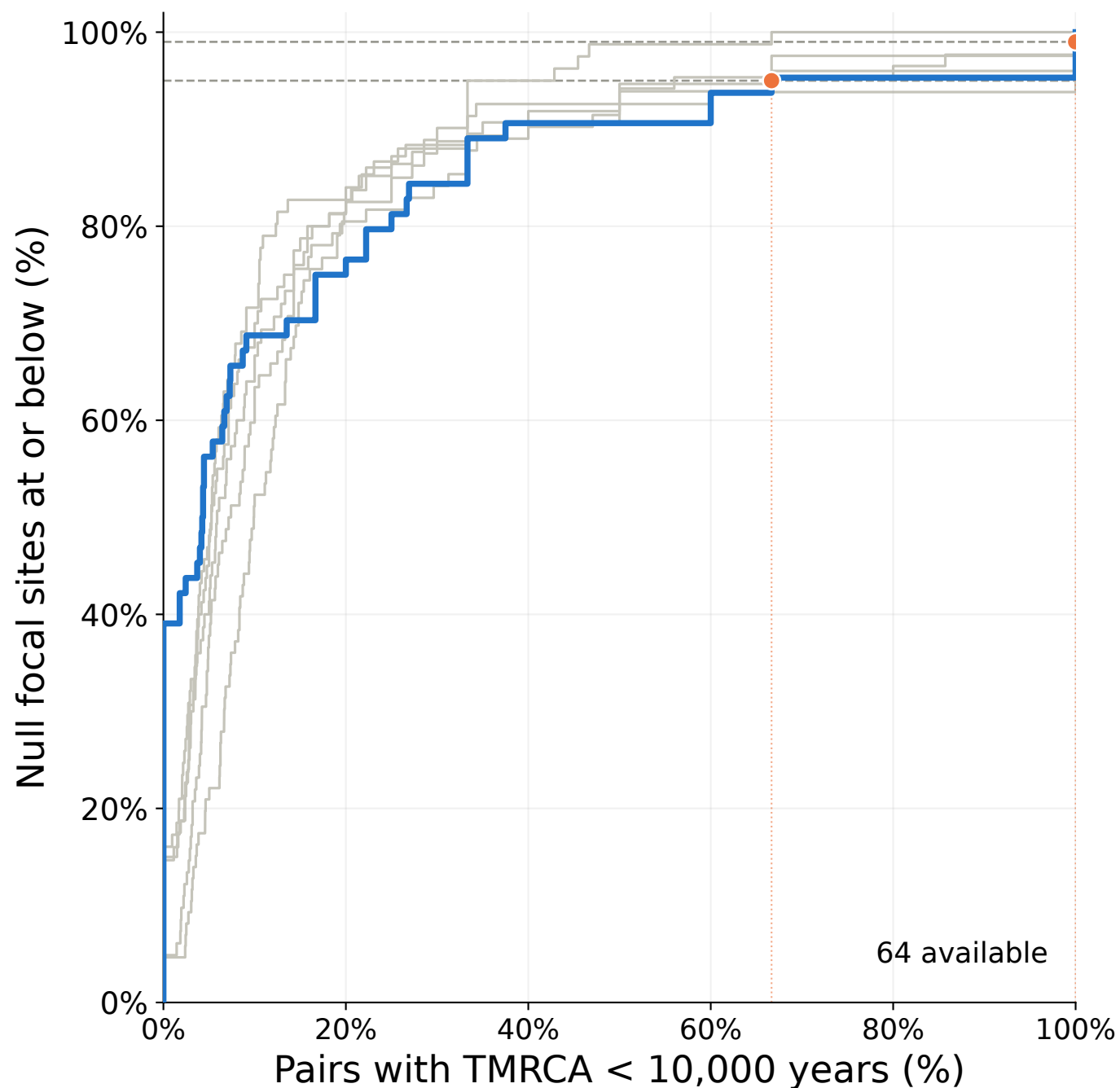


PROVISIONAL completed subset | step CDF; descriptive empirical percentiles
Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.

Truth TMRCA | ALT/ALT pairs

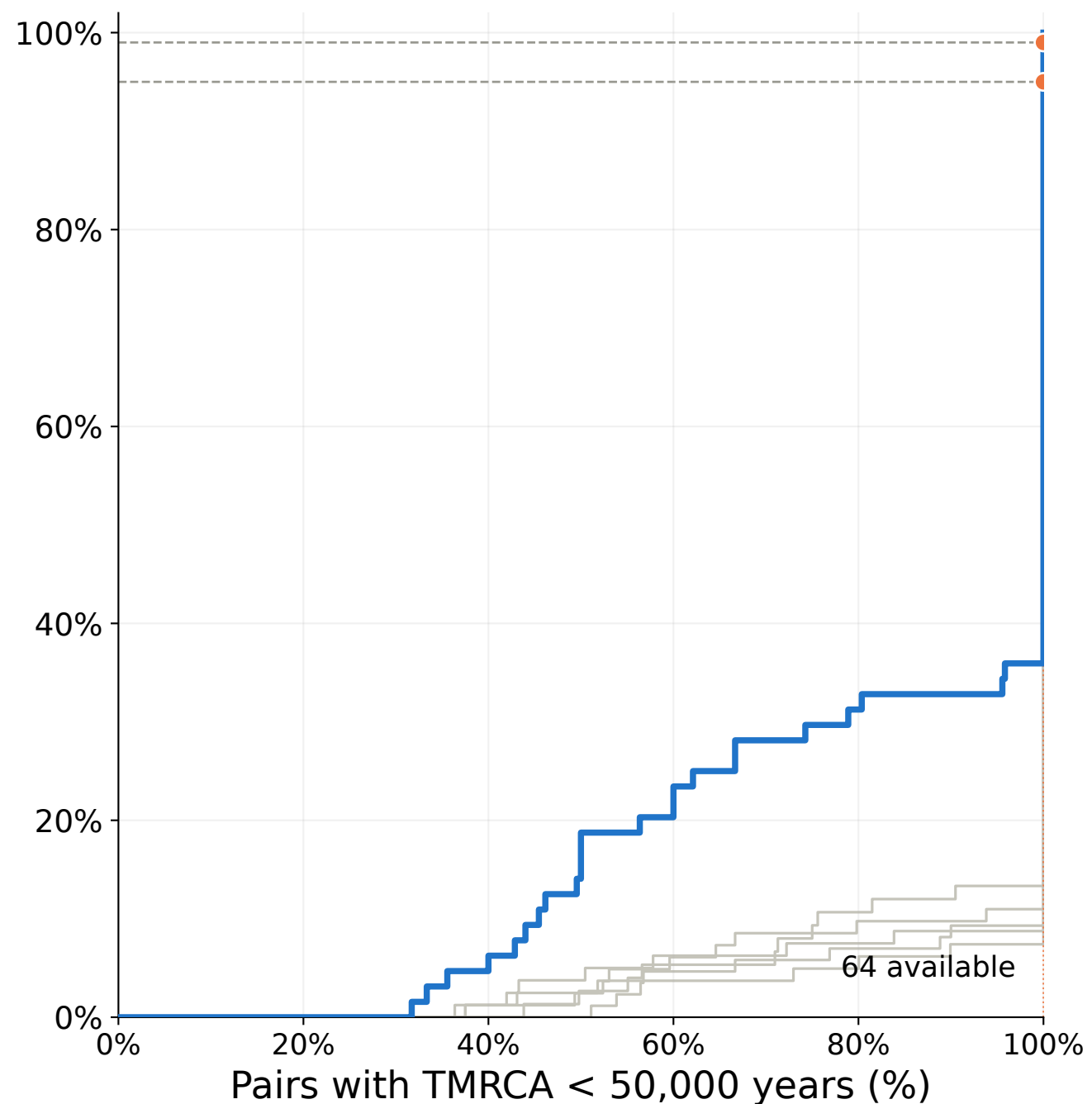
AFR | n = 88

Top 5%: 66.67% Top 1%: 100.00%



AFR | n = 88

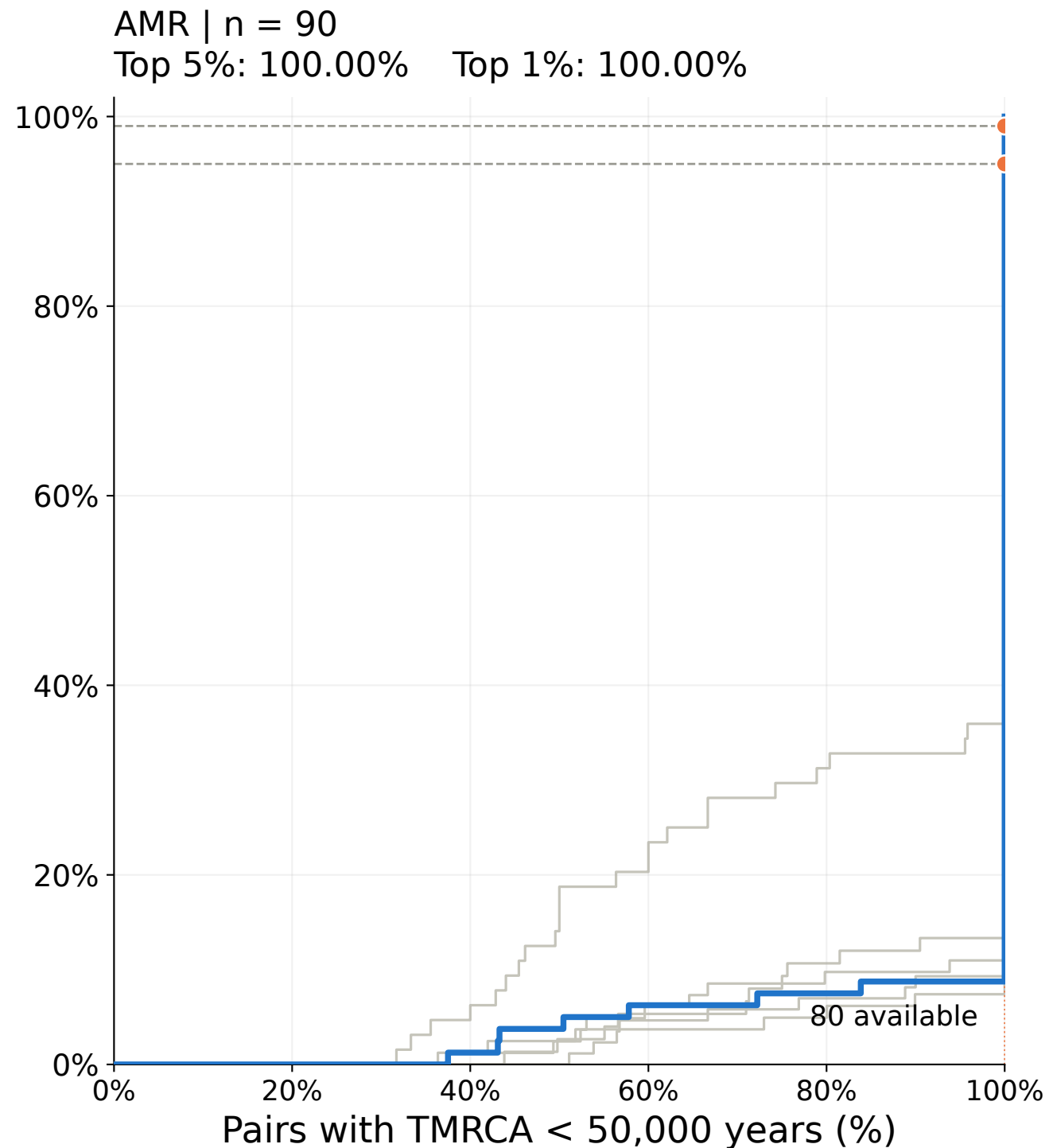
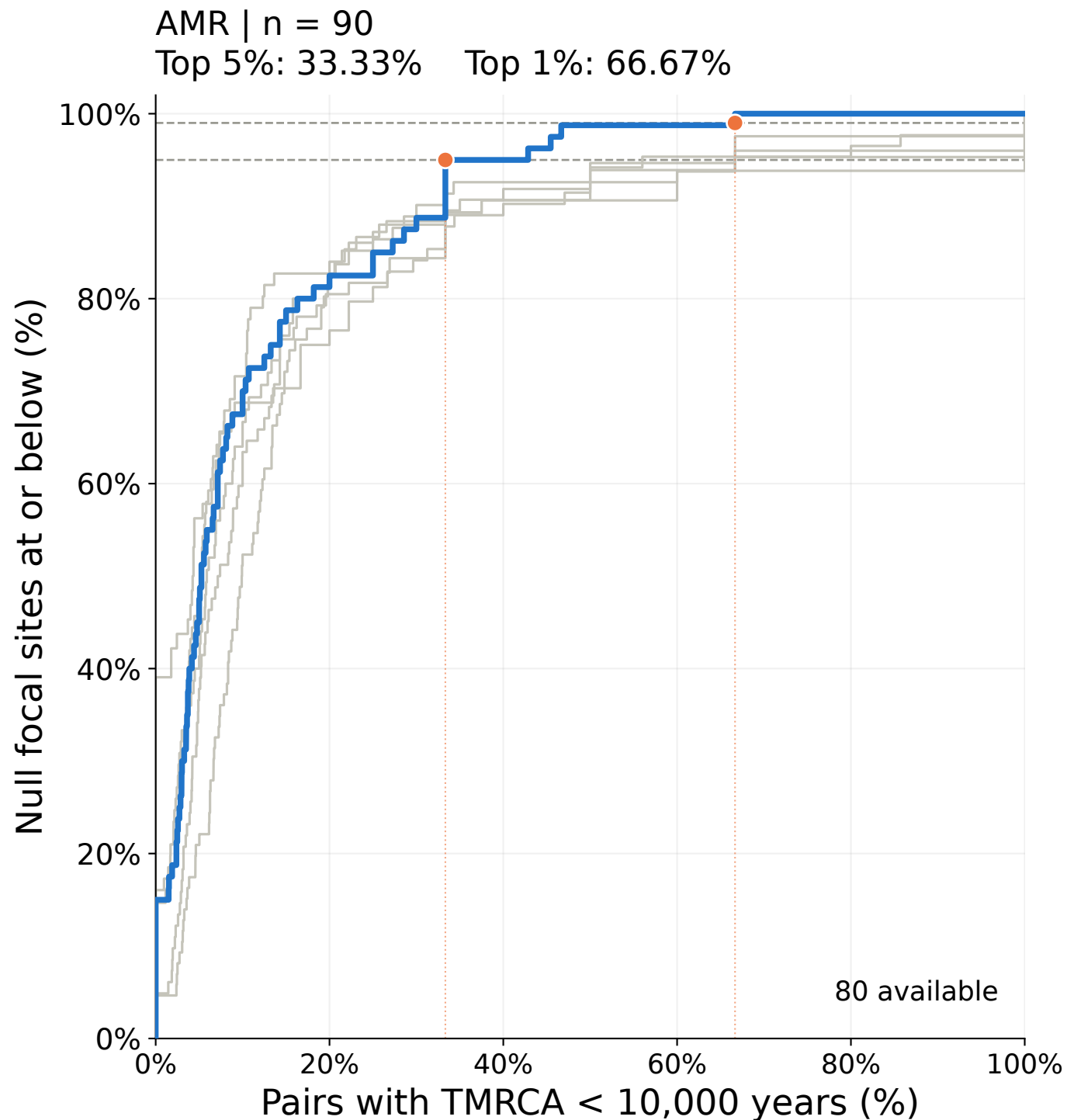
Top 5%: 100.00% Top 1%: 100.00%



PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

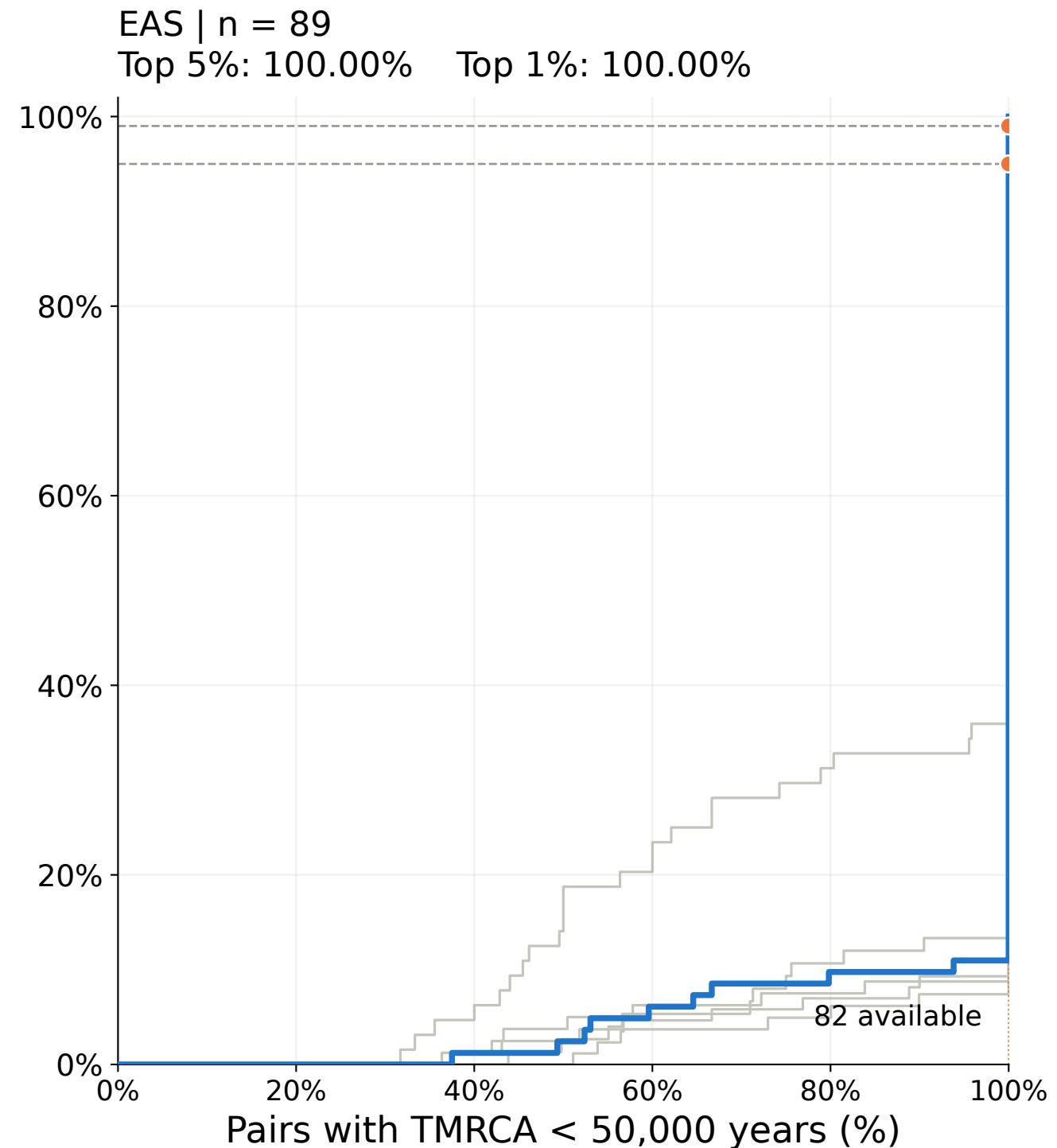
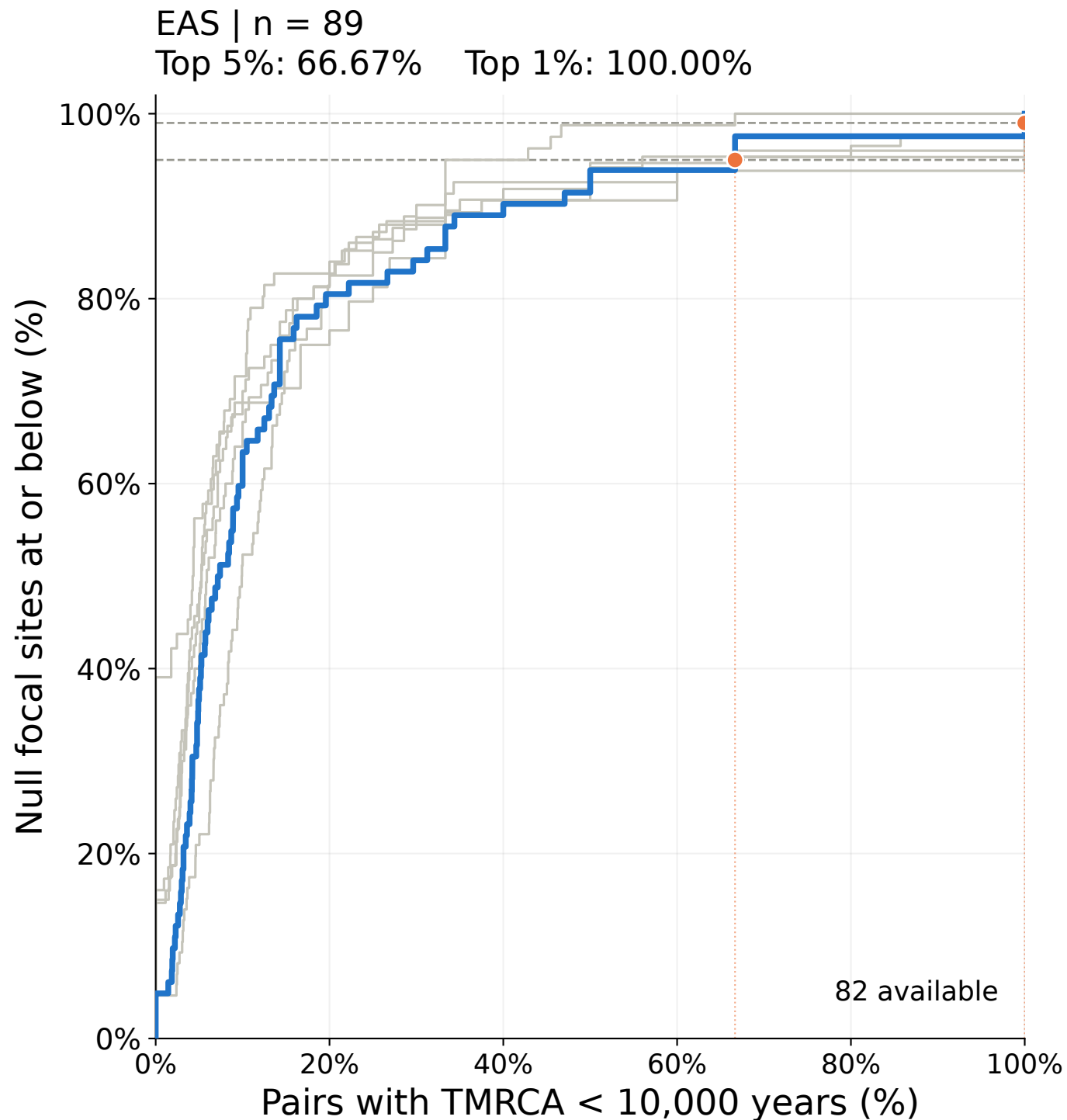
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Truth TMRCA | ALT/ALT pairs



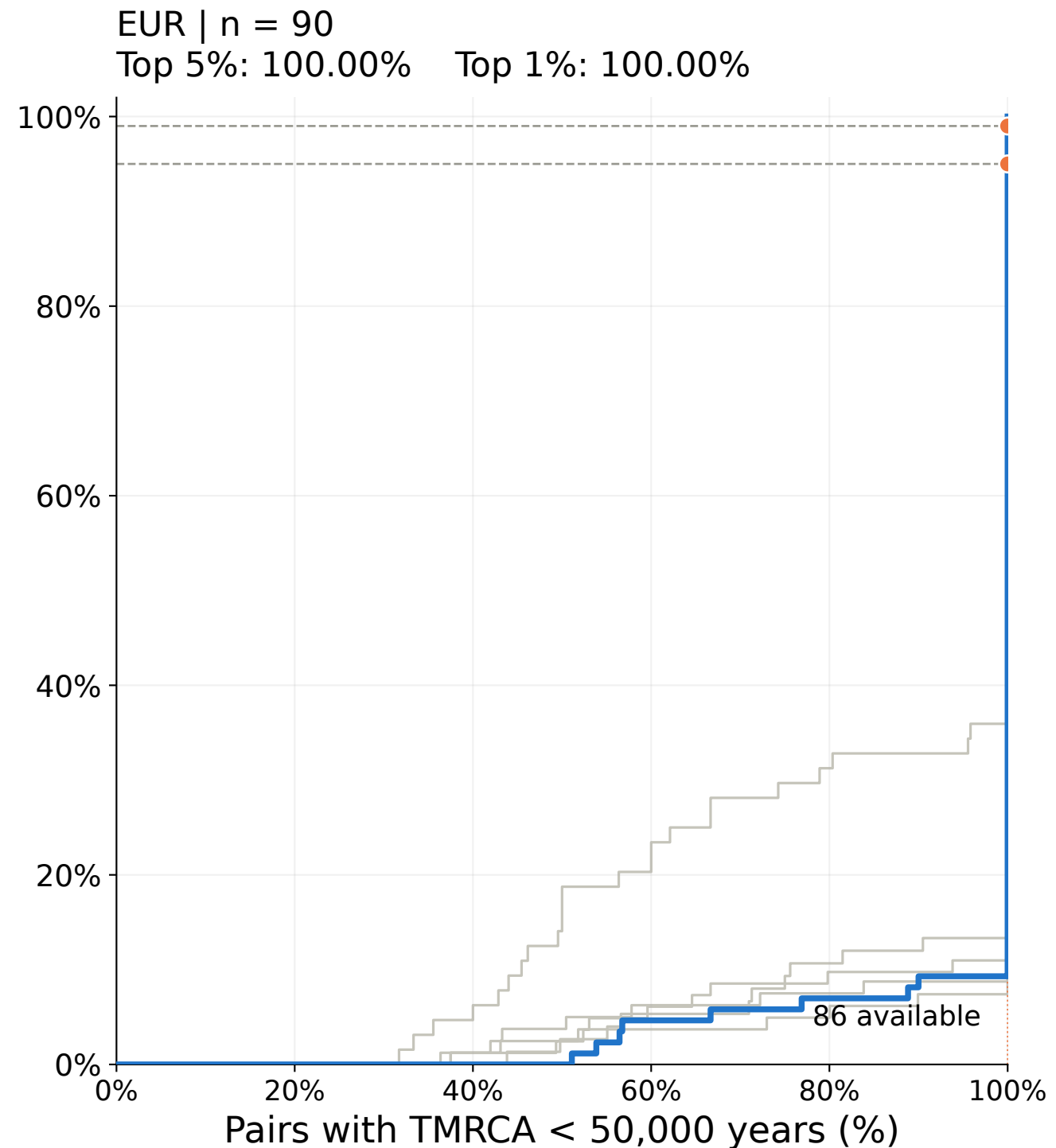
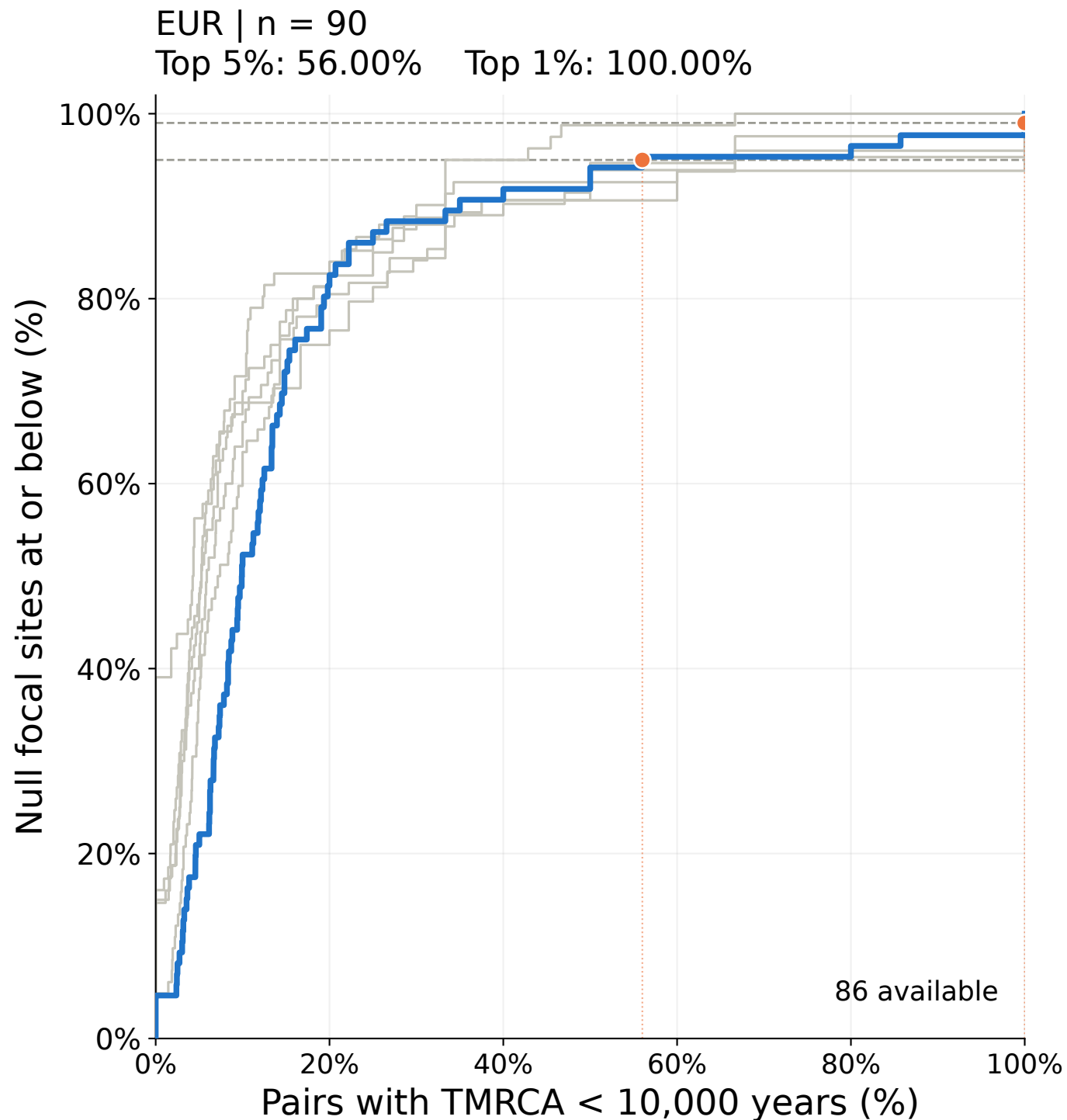
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Truth TMRCA | ALT/ALT pairs



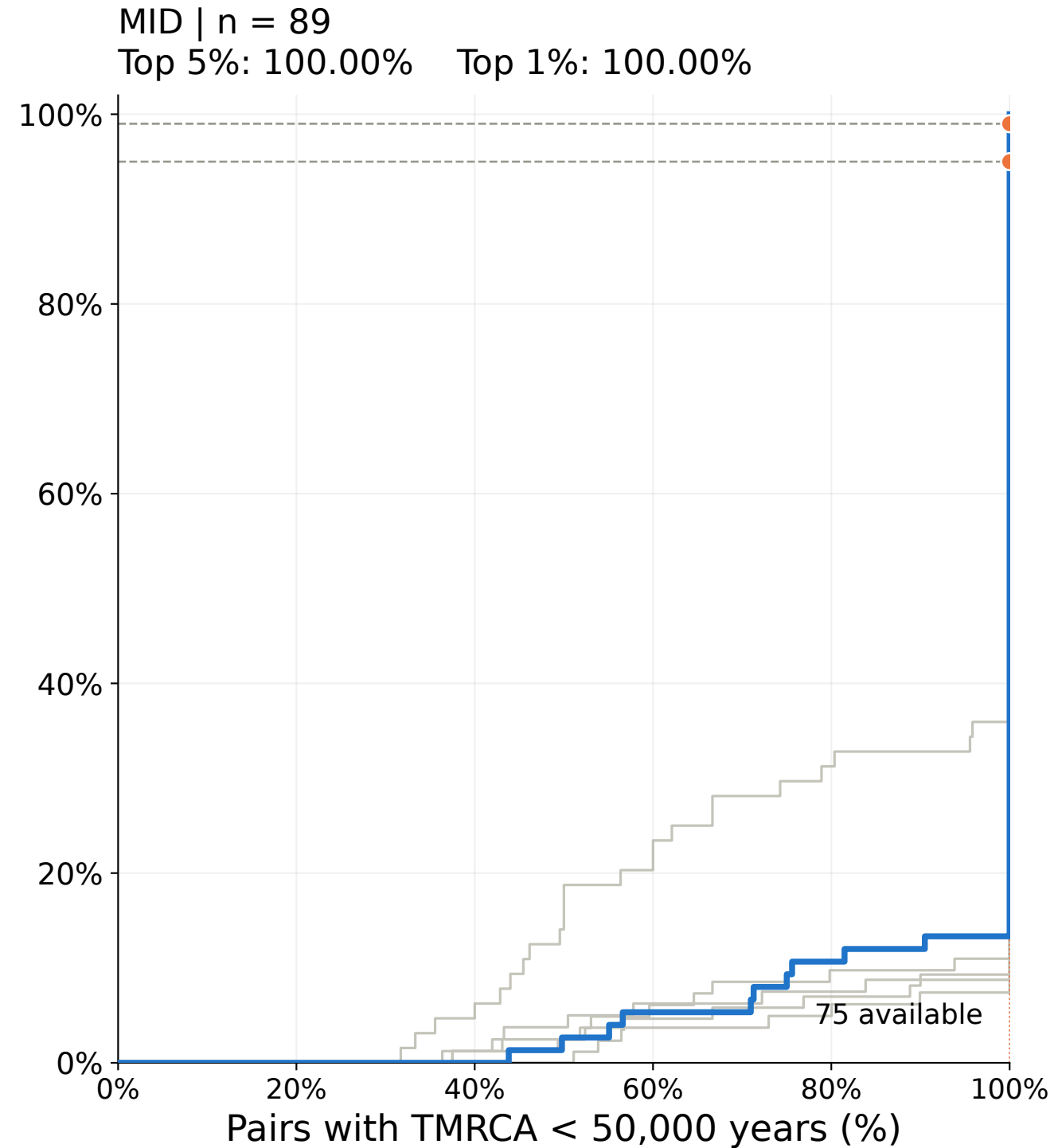
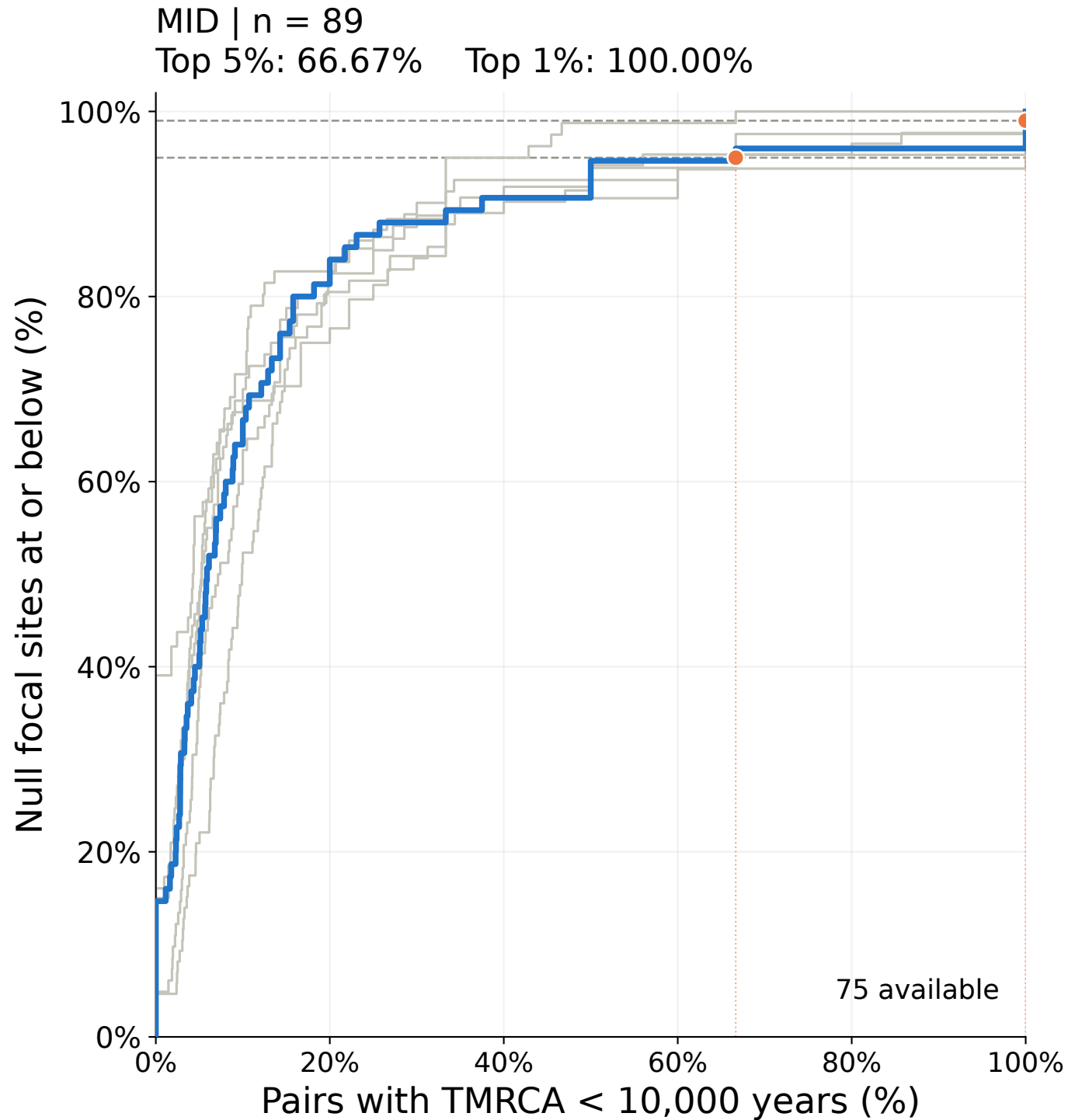
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Truth TMRCA | ALT/ALT pairs



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Truth TMRCA | ALT/ALT pairs

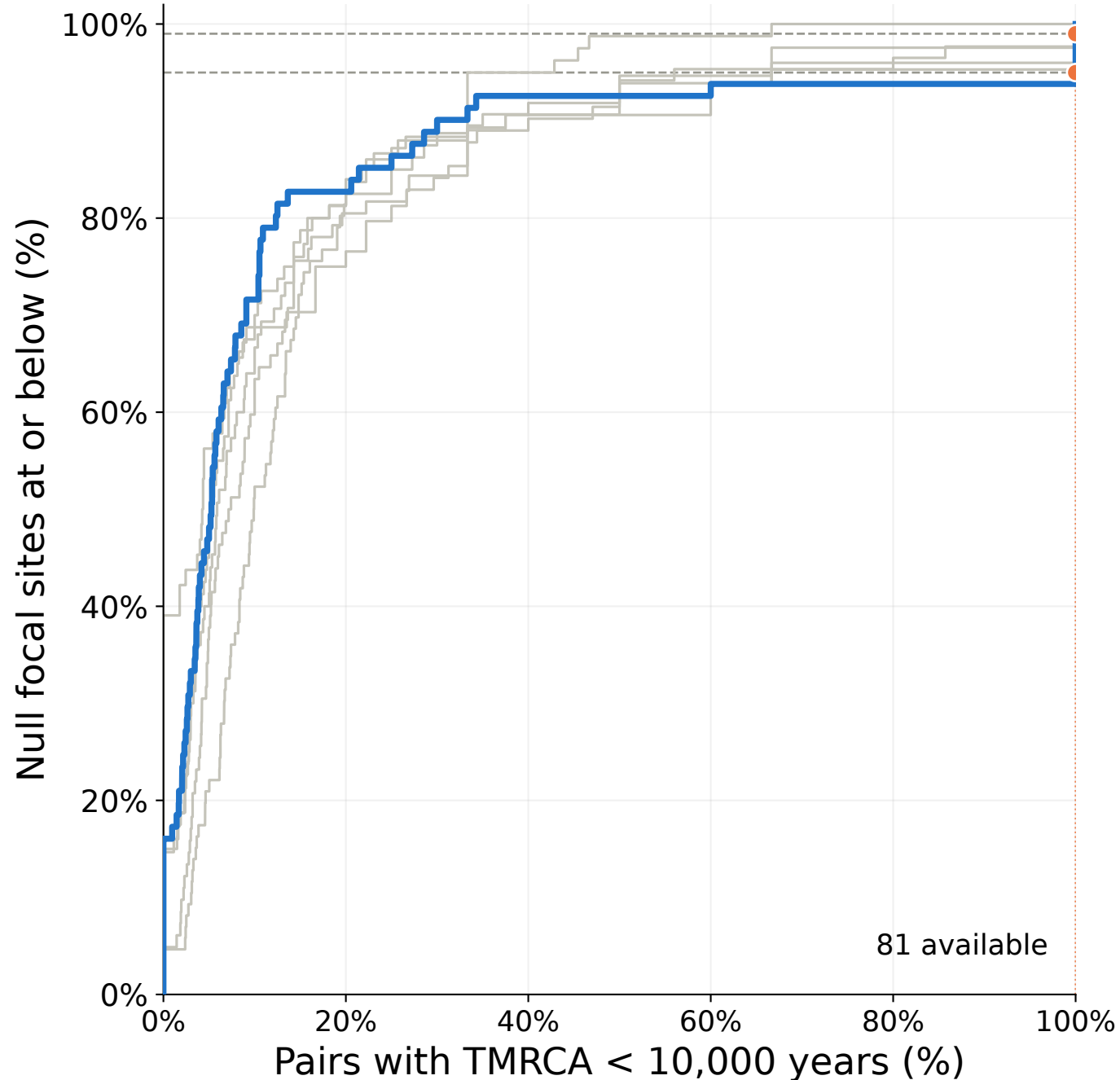


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Truth TMRCA | ALT/ALT pairs

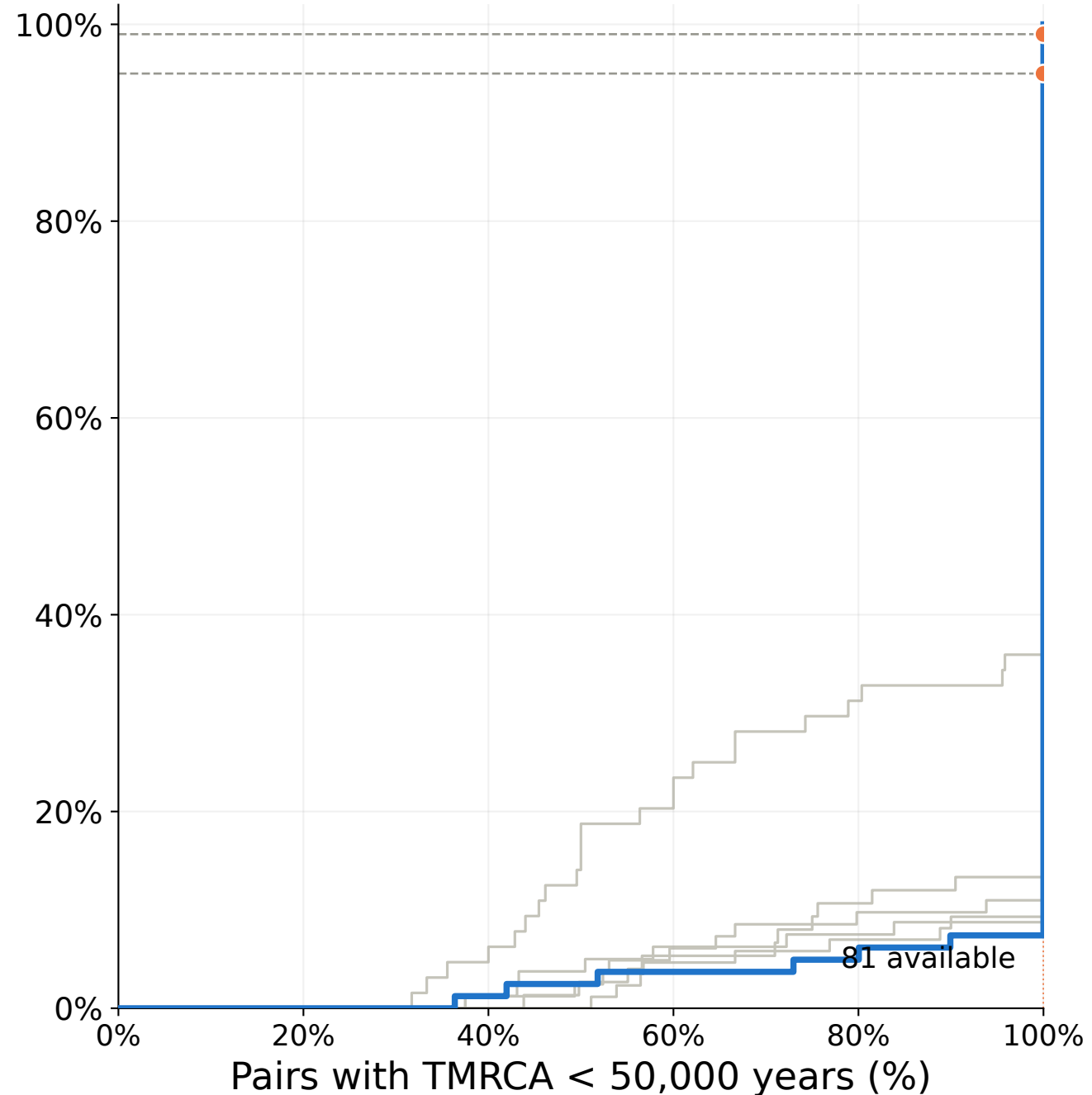
SAS | n = 89

Top 5%: 100.00% Top 1%: 100.00%



SAS | n = 89

Top 5%: 100.00% Top 1%: 100.00%



PROVISIONAL completed subset | step CDF; descriptive empirical percentiles

Blue: focal population; gray: other populations. Percentiles are not finite-null p-value boundaries.